

The M-CATs

Biochemistry Chapter 9-10: PDH, Acetyl-CoA, Citric Acid Cycle, PPP

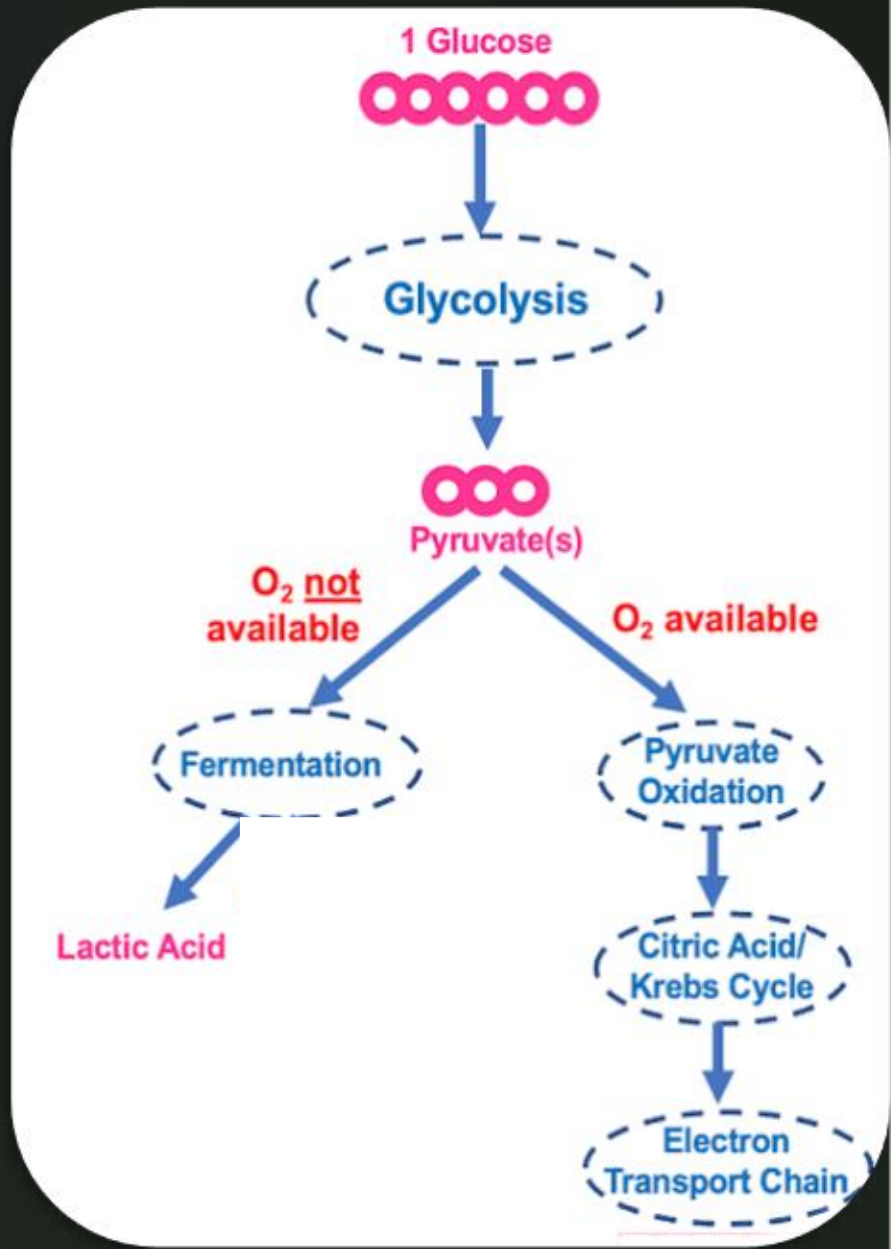
PDH and Citric Acid Cycle Basics

We just talked about glycolysis and lactic acid fermentation, which breaks down glucose (but can be reversed)

Pyruvate can instead be further oxidized for WAY more ATP (28-32 ATP).

These steps are irreversible; the product CO2 cannot be rejoined to acetyl-CoA to form pyruvate (unlike glucose regeneration in Cori cycle)

These pathways/stages take place in the mitochondria, unlike cytosolic glycolysis and lactate fermentation

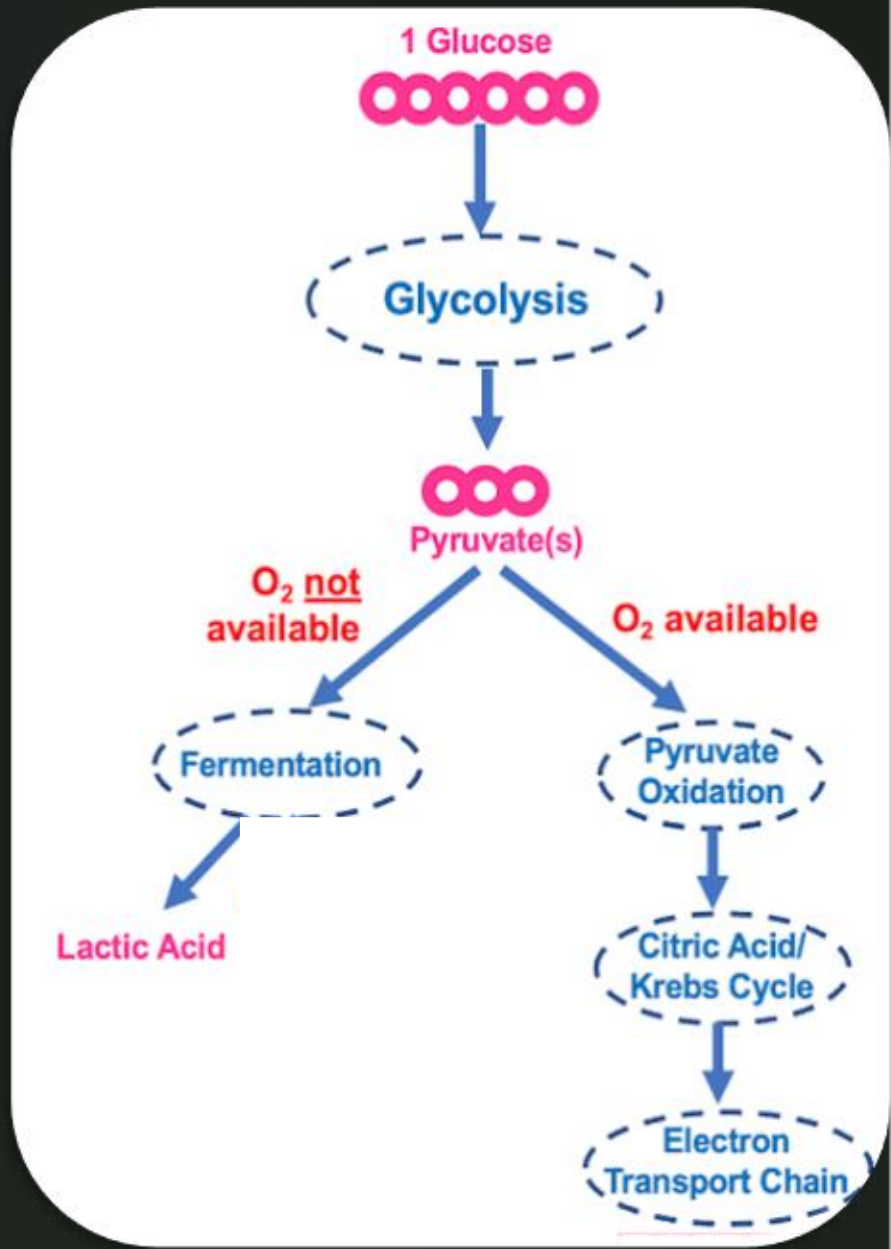


PDH and Citric Acid Cycle Oxidation

These two steps work together to COMPLETELY oxidize all carbons from the original glucose into CO2

We will see a lot of "Dehydrogenases" in these pathways. These are enzymes that remove $2H^+$ and $2e^-$ from a molecule and transfer them to a coenzyme.

In effect, these dehydrogenases oxidize products of glucose

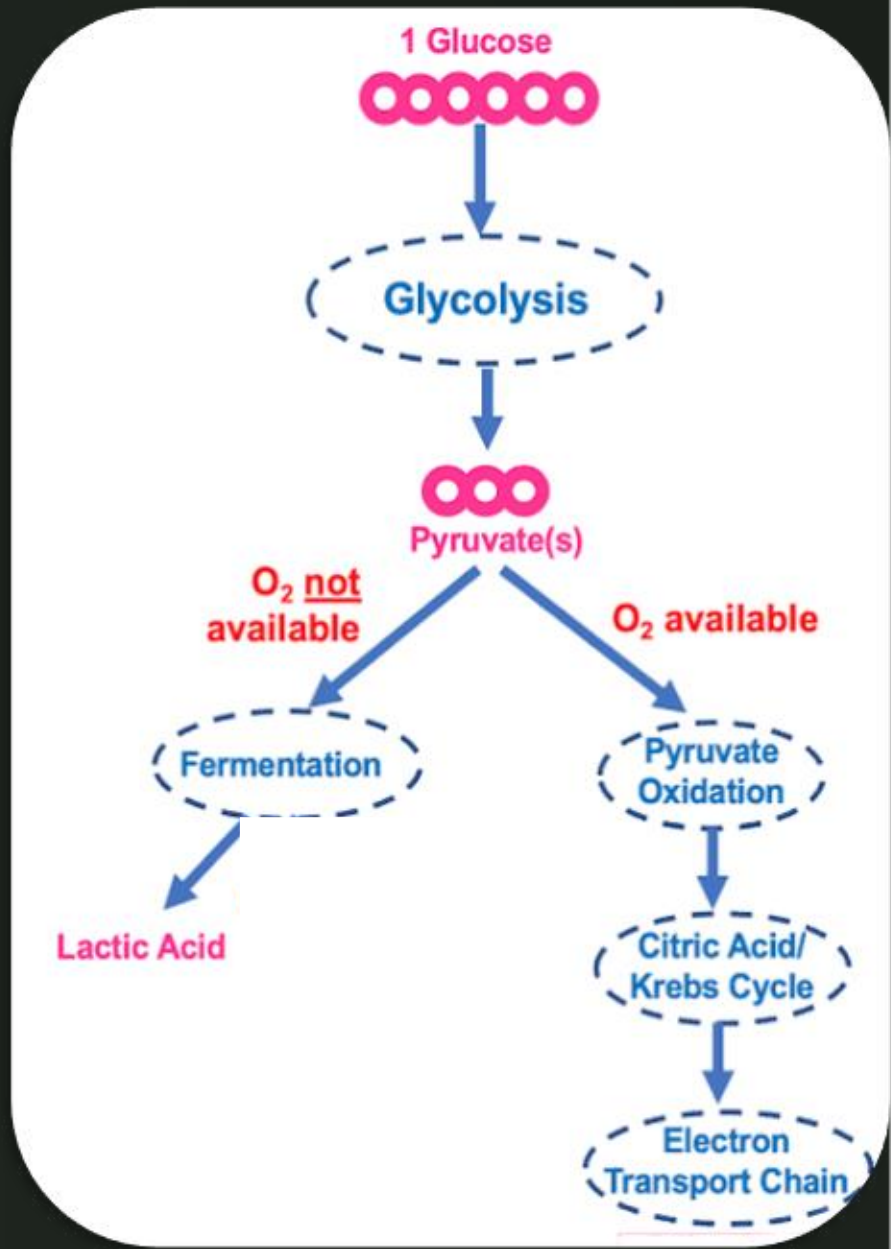


PDH and Citric Acid Cycle Basics

Pyruvate enters the mitochondria and is broken down to 3 CO2.

The net result is production of electron carriers = reducing power = NADH and FADH2 (which will be used/cashed in later)

We also get some ATP and GTP out of the process (chump change)



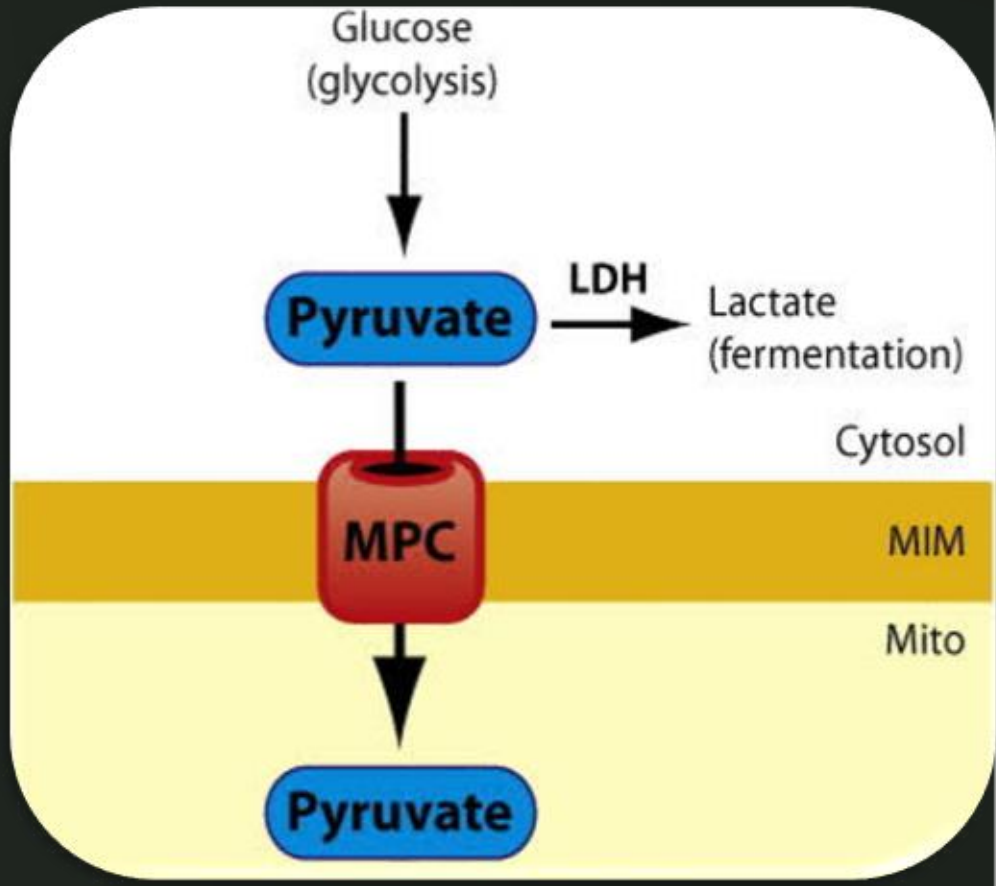
PDH and Acetyl-CoA

Pyruvate enters Mitochondrial Matrix

Pyruvate diffuses into intermembrane space, then is transported into the mitochondrial matrix by the Mitochondrial Pyruvate Carrier

MPC is a secondary active transporter - symporter

1 H+ and 1 pyruvate are transported from the intermembrane space to the mitochondrial matrix

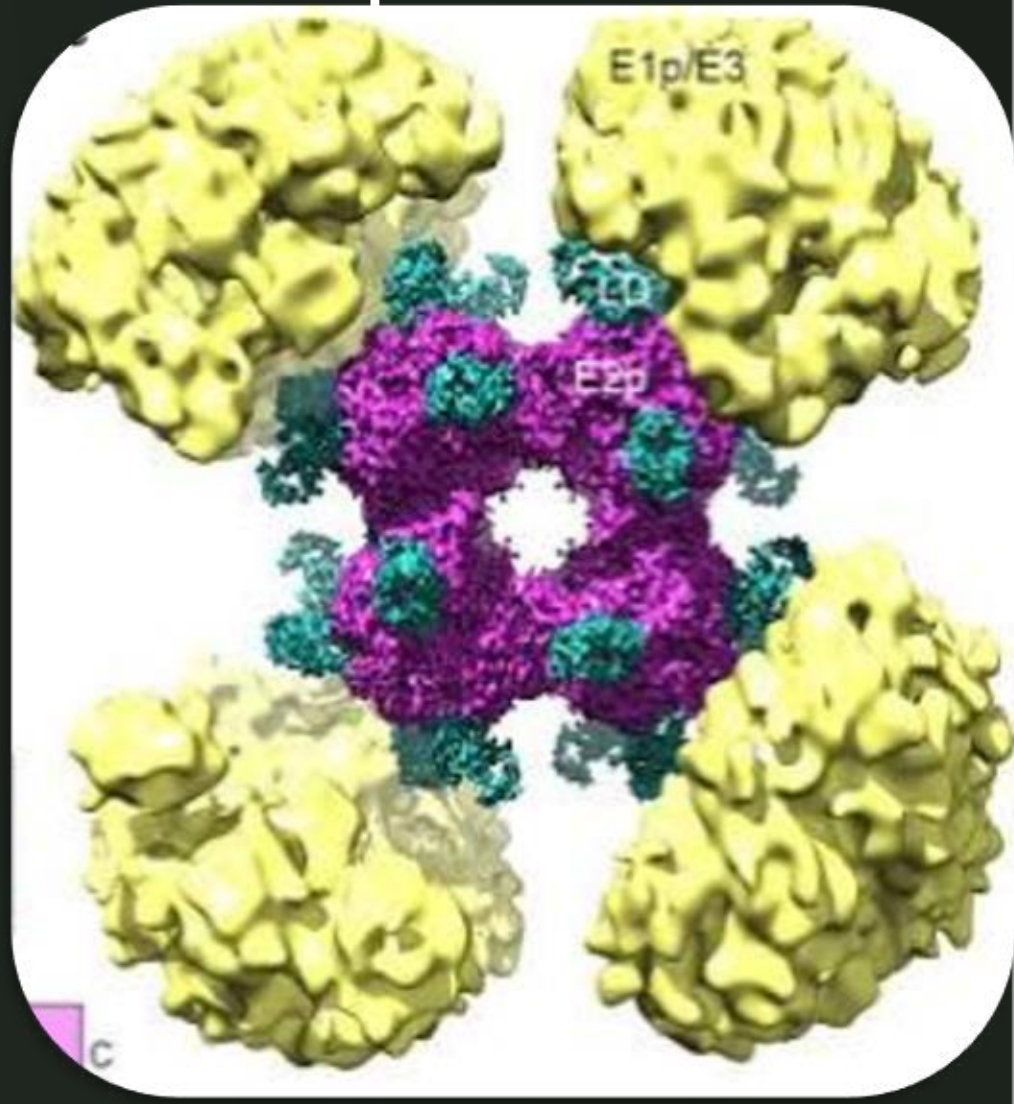


Pyruvate Dehydrogenase Complex

The PDH complex is composed of 5 total enzymes. 3 are used to oxidize pyruvate into acetyl CoA: pyruvate dehydrogenase (PDH), dihydrolipoyl transacetylase, and dihydrolipoyl dehydrogenase

The other two enzymes, pyruvate dehydrogenase kinase and pyruvate dehydrogenase phosphatase, are regulatory

5 coenzymes are utilized by the complex: thiamine pyrophosphate, lipoic acid, CoA, FAD, and NAD+

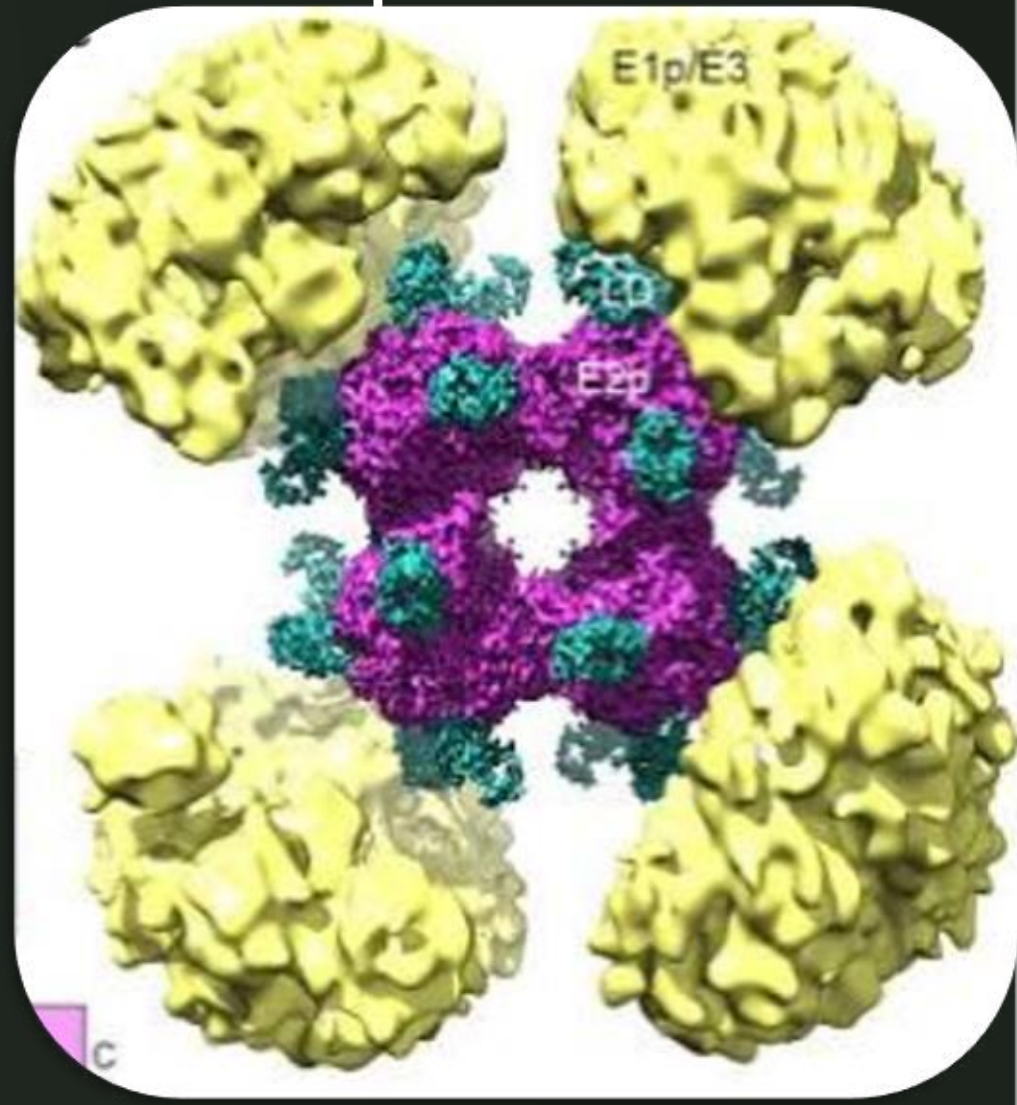


Pyruvate Dehydrogenase Complex

PDH complex has enzymes all together for efficiency:

Any intermediate can immediately enter the next enzyme in the process without diffusion delay. This is important when the reaction is a set of group transfers from one coenzyme to another

One coenzyme literally acts as an arm that swings from one enzyme to another, facilitating movement of intermediates

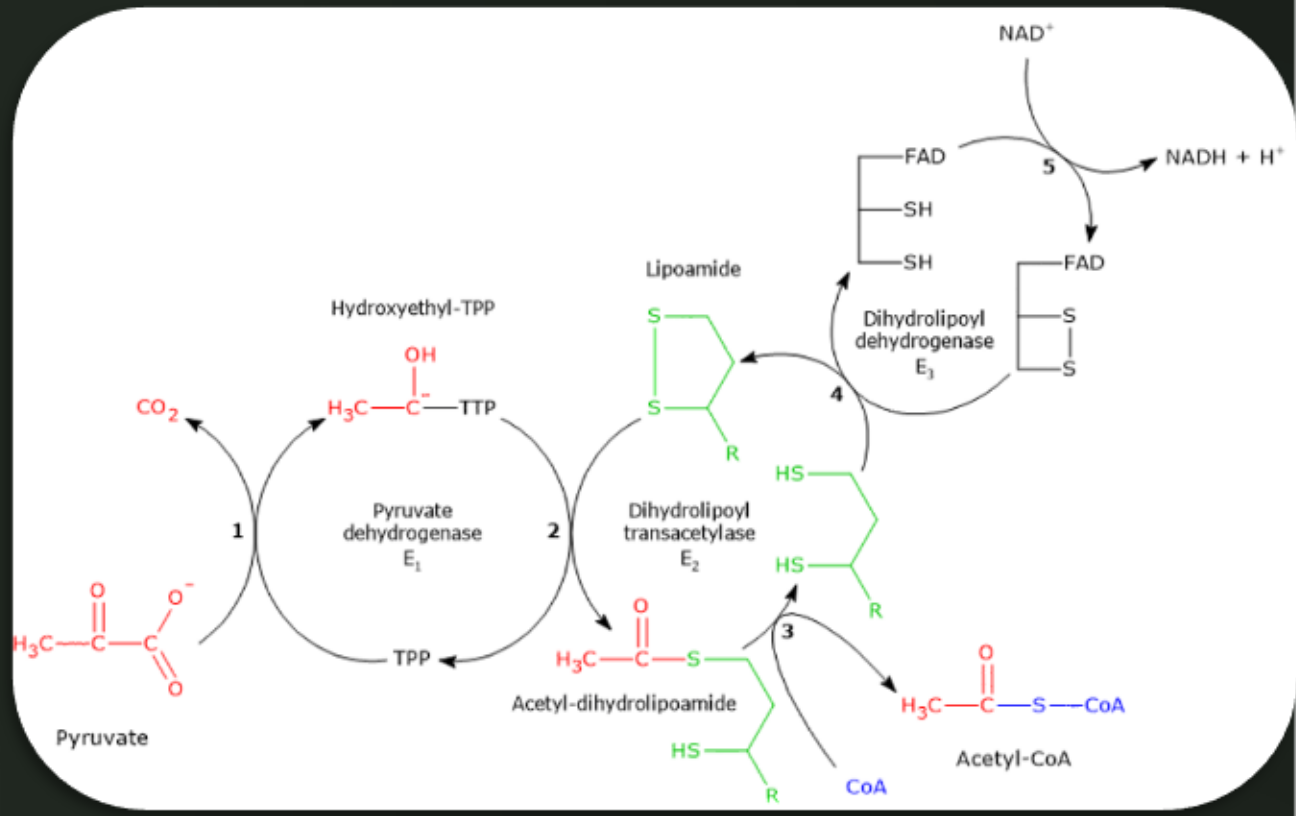


PDH Complex Reactions

Pyruvate Dehydrogenase and Dihydrolipoyl Transacetylase pass the acetyl group around

Acetyl group is eventually passed to CoA, forming Acetyl-CoA (and byproduct CO_2)

Dihydrolipoyl Dehydrogenase then restores the disulfide bond by reducing FAD^+ to $FADH_2$, then reducing NAD^+ to NADH to restore the FAD^+ coenzyme.



PDH Kinase and PDH Phosphatase

These are the last two enzymes of the PDH complex. Both of these enzymes are bound to the complex

These enzymes phosphorylate and dephosphorylate PDH, respectively, changing the activity of PDH. Phosphorylation by the Kinase *inactivates* PDH.

PDH Kinase and PDH Phosphatase receive signals relating to energy balance; high NADH/NAD+, ATP/ADP, and acetyl-CoA/CoA-SH will stimulate the former to inactivate PDH.

The opposite signals, in addition to Ca^{2+} and some hormones (depending on the tissue), stimulate PDH phosphatase to activate PDH



Acetyl-CoA is Central to Central Metabolism

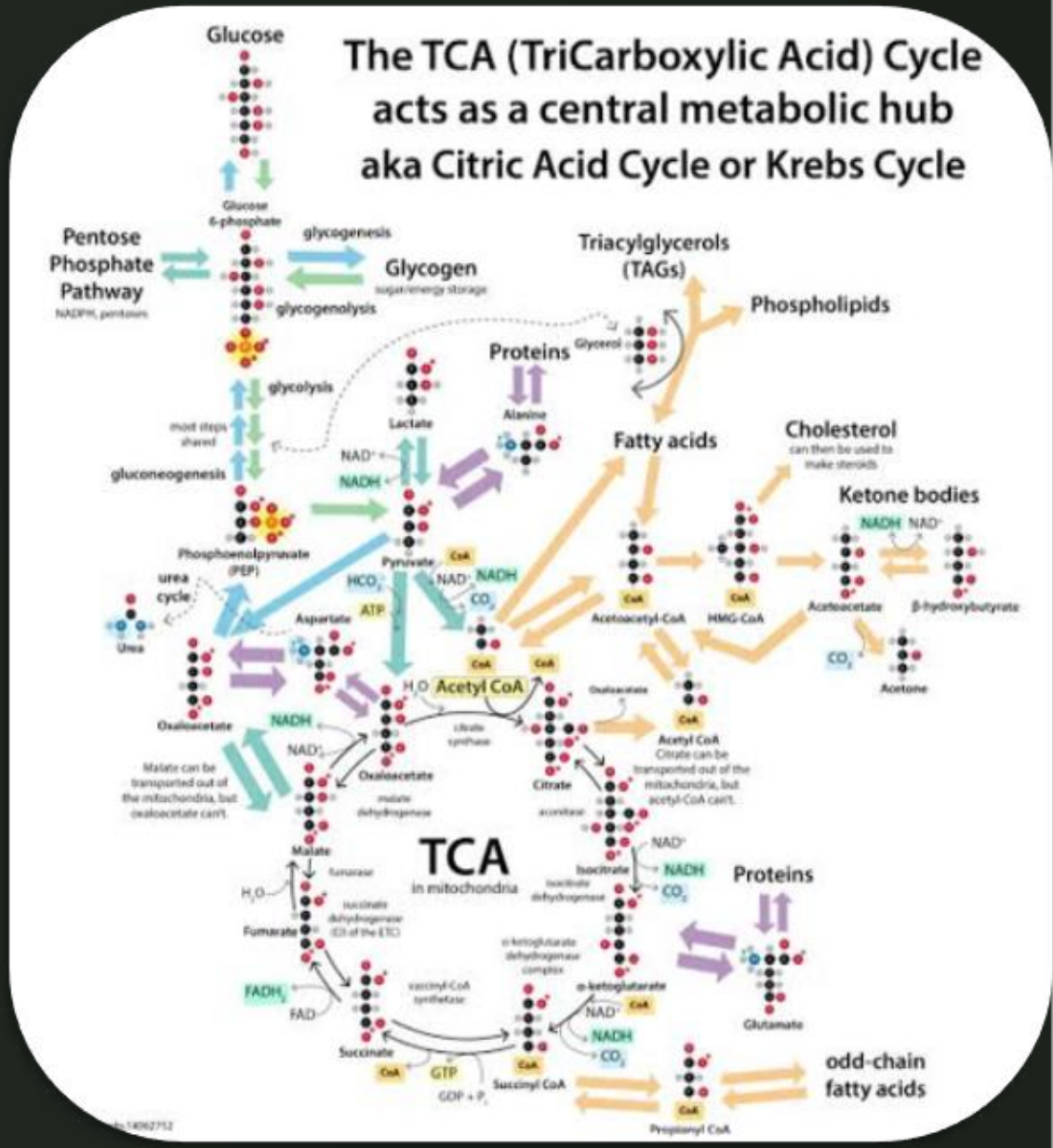
We already know that Acetyl-CoA is produced as an intermediate of complete glucose oxidation.

Acetyl-CoA is produced during fat breakdown (beta-oxidation) and is used to form new fatty acids.

Ketogenic Amino Acids can be broken down to yield acetyl-CoA.

Ketone Bodies are made from and can be converted into acetyl-CoA.

Acetyl-CoA can be fully oxidized to CO_2. One acetyl-CoA is worth 10-12 ATP.

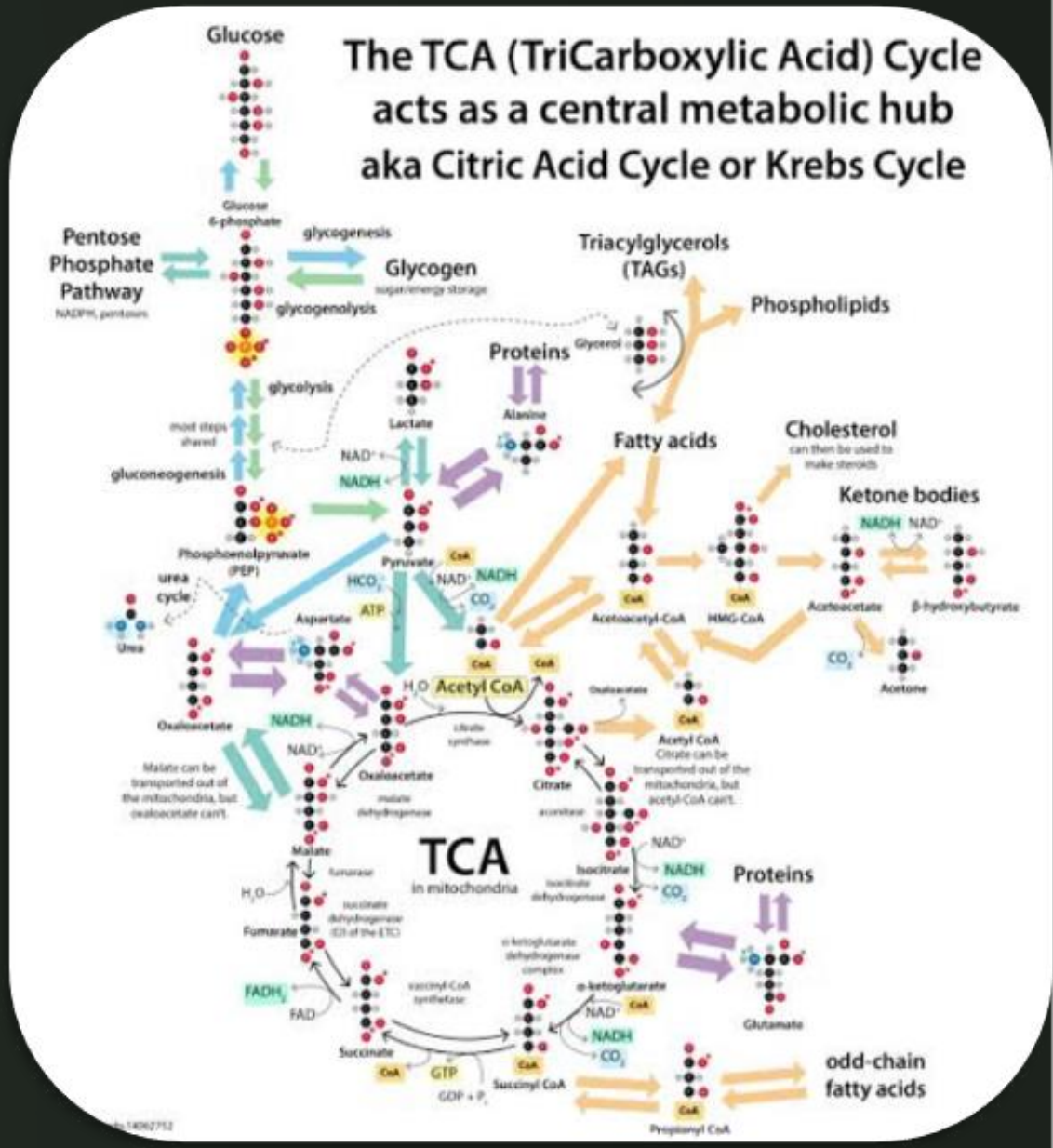


Acetyl-CoA is Central to Central Metabolism

When alcohol is consumed, alcohol dehydrogenase and acetaldehyde dehydrogenase convert it to acetyl-CoA.

Unfortunately, NADH is produces in this reaction, which inhibits the TCA cycle (there is no need for the cycle when NADH is abundant).

As a result, the Acetyl-CoA produced is almost entirely converted into fatty acids.



Citric Acid Cycle (TCA)

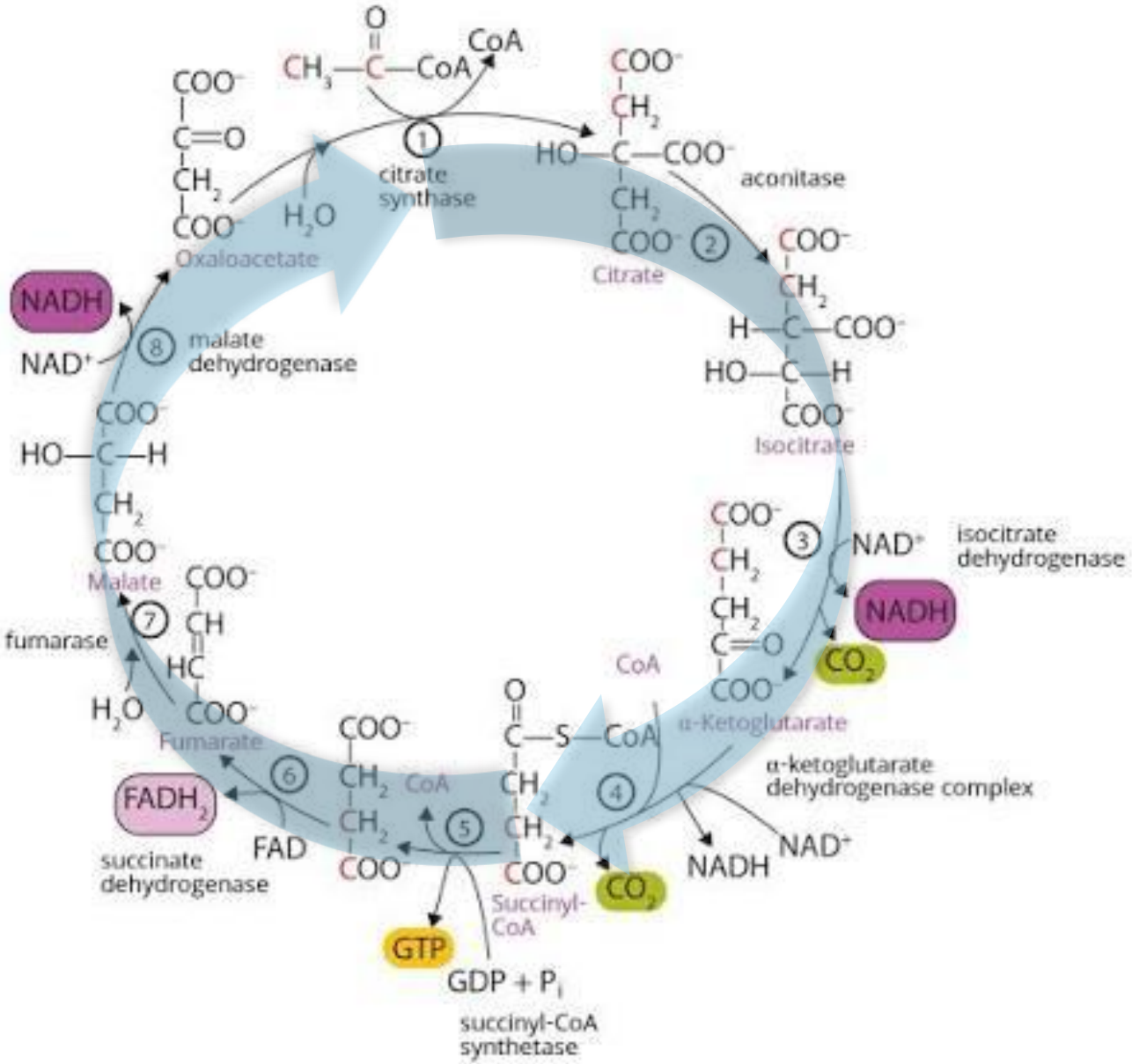
Citric Acid Cycle Overview

This cycle is like a larger, expanded Pyruvate Oxidation process:
CO2 is released, and electron carriers are produced as the important product.

Steps 1 through 4 accomplish CO2 release (twice, equal to a full acetyl-CoA)

Steps 5 through 8 regenerate molecules used in the beginning of the cycle; in doing so, they generate electron carriers

Step 4's complex is homologous to PDH complex, and also attaches CoA to an intermediate, just like the PDH complex

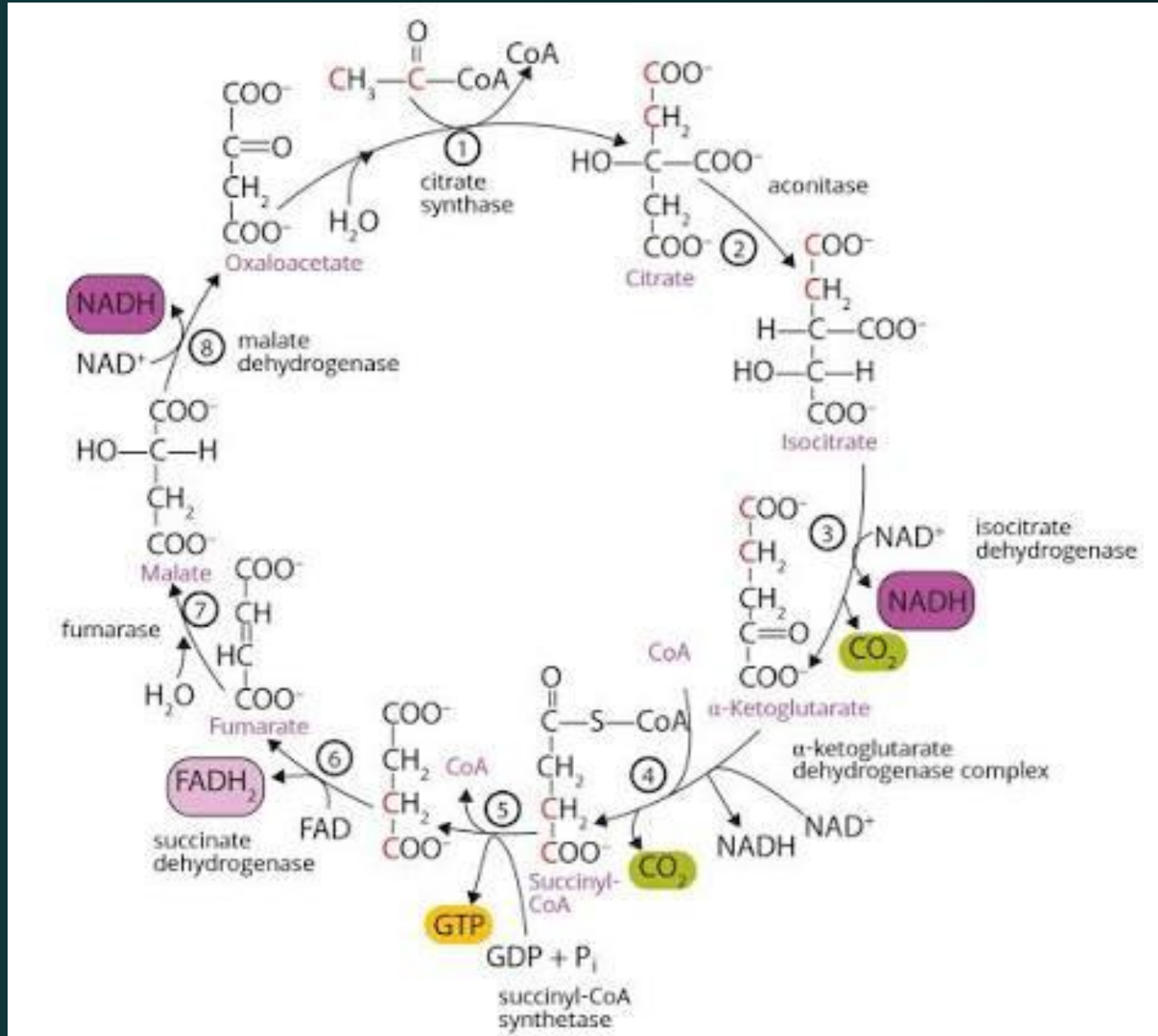


Citric Acid Cycle Naming

All enzymes are named after the precursor/substrate that they act on, except citrate synthase (which **produces** citrate) and aconitase (which converts citrate to isocitrate, isomerization)

Intermediates, in order, fit acronym
Can I Keep Selling Sex For Money, Officer?

All enzymes named “dehydrogenase” will oxidize an intermediate and produce an electron carrier (NADH or $FADH_2$) during their step.



Citric Acid Cycle Carbon Tracking

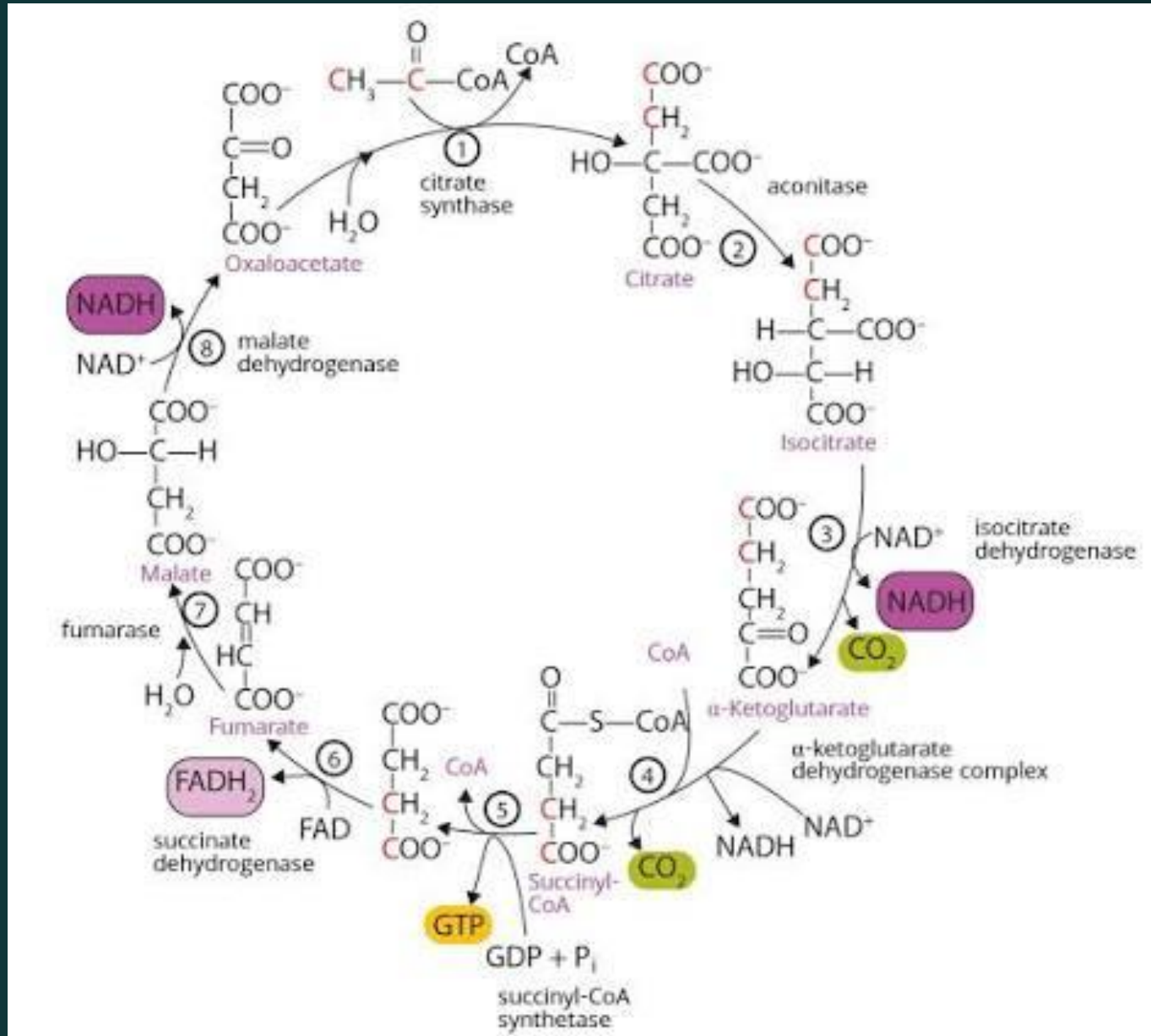
Acetyl-CoA enters at stage 1, forming 6-carbon citrate.

At stage 3, isocitrate dehydrogenase removes one CO_2, producing 5-carbon alpha-ketoglutarate.

At stage 4, alpha-ketoglutarate dehydrogenase complex removes a second CO_2, producing 4-carbon Succinyl-CoA.

After stage 5 removes CoA, the next 3 stages just rearrange on a 4-carbon skeleton

The CO_2 produced aren't from acetyl-CoA this cycle: rearrangement occurs



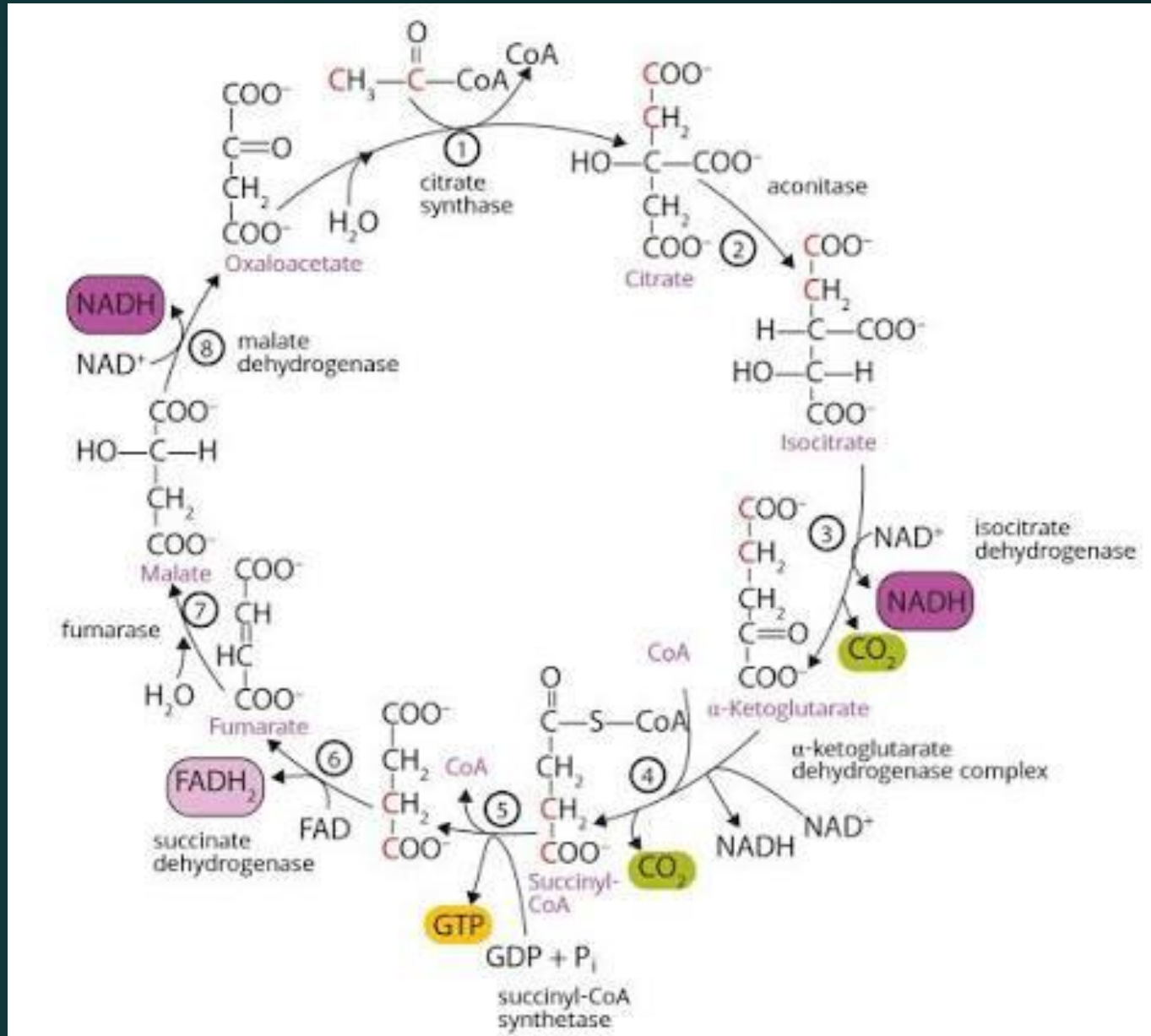
Citric Acid Cycle Carbon Tracking

NADH is produced after KIM: alpha-ketoglutarate, isocitrate, and malate.

CoA is added to produce Succinyl CoA

CoA is removed in the same step that GTP is formed (right after CoA is added – you only see “CoA” in the intermediate names once…)

FAD is reduced to $FADH_2$ by succinate dehydrogenase, the only dehydrogenase left. This enzyme is a part of the Electron Transport Chain (It IS Complex II)



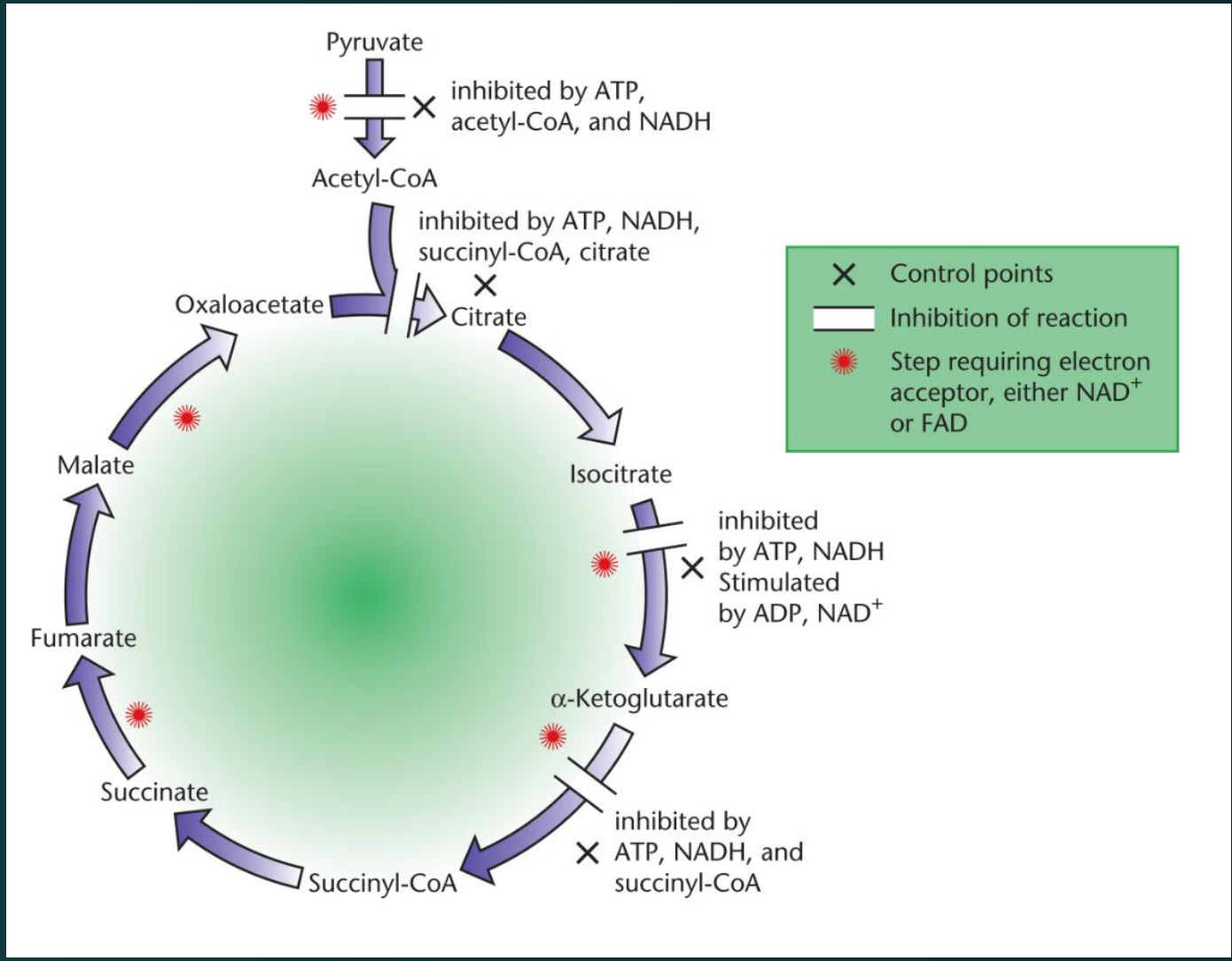
Citric Acid Cycle Regulation

The cycle is controlled by metabolites acting as inhibitors and activators

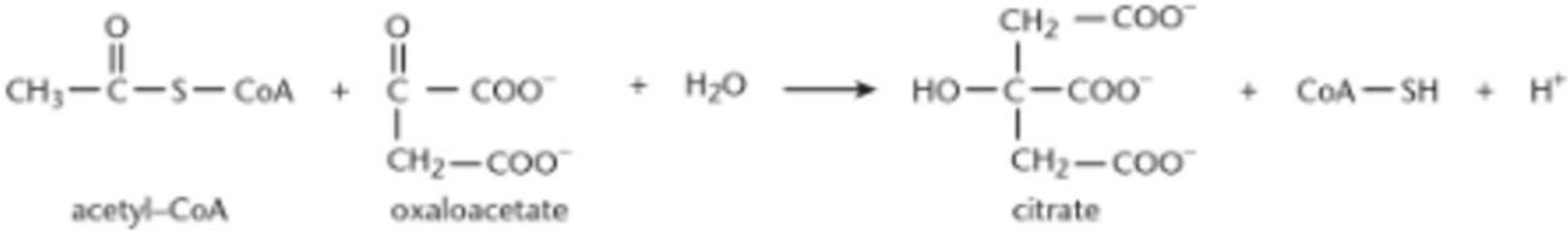
Isocitrate Dehydrogenase, Alpha-Ketoglutarate Dehydrogenase, and Citrate Synthase are regulated enzyme

The products of the cycle (NADH, ATP) act as inhibitors. This is called feedback inhibition/negative feedback.

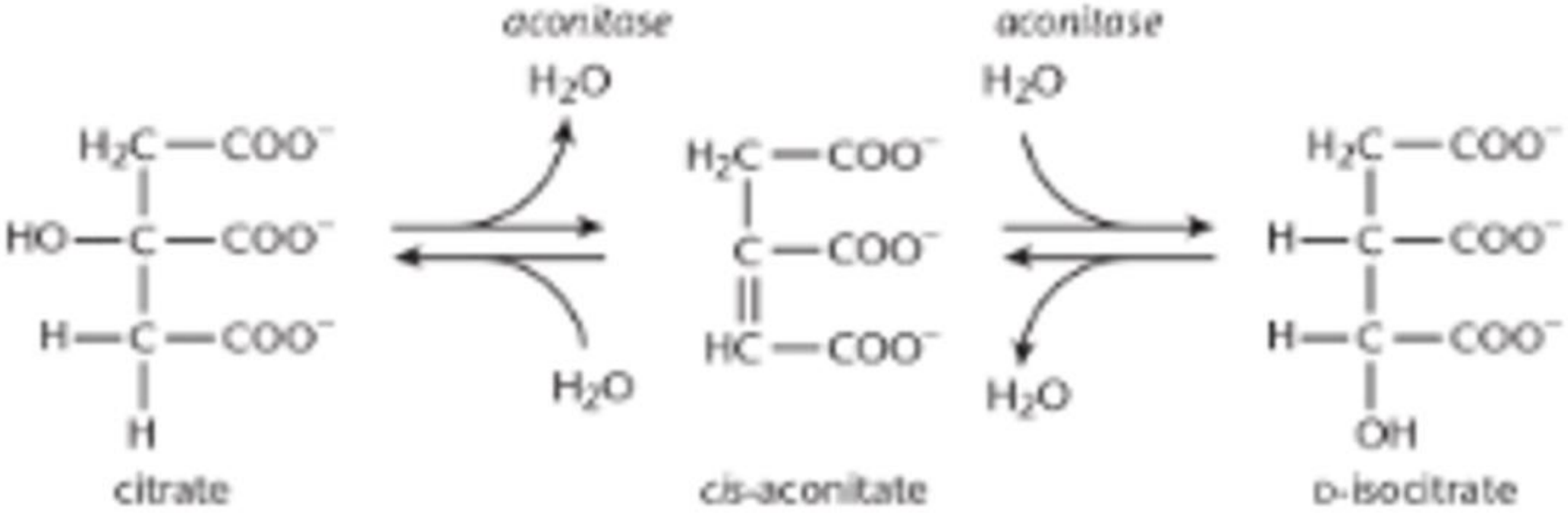
Ca2+, NAD+, ADP stimulate isocitrate dehydrogenase (all of these should signal energy need)



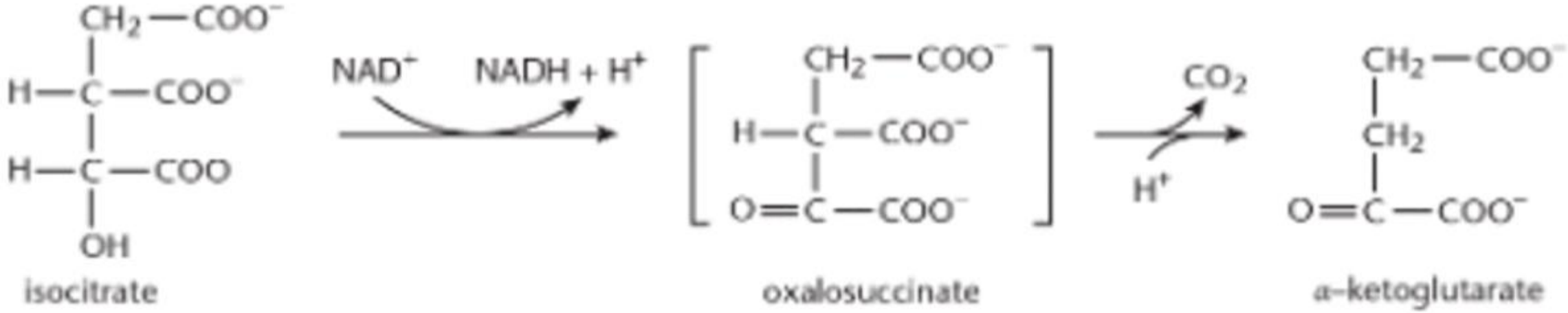
Citric Acid Cycle



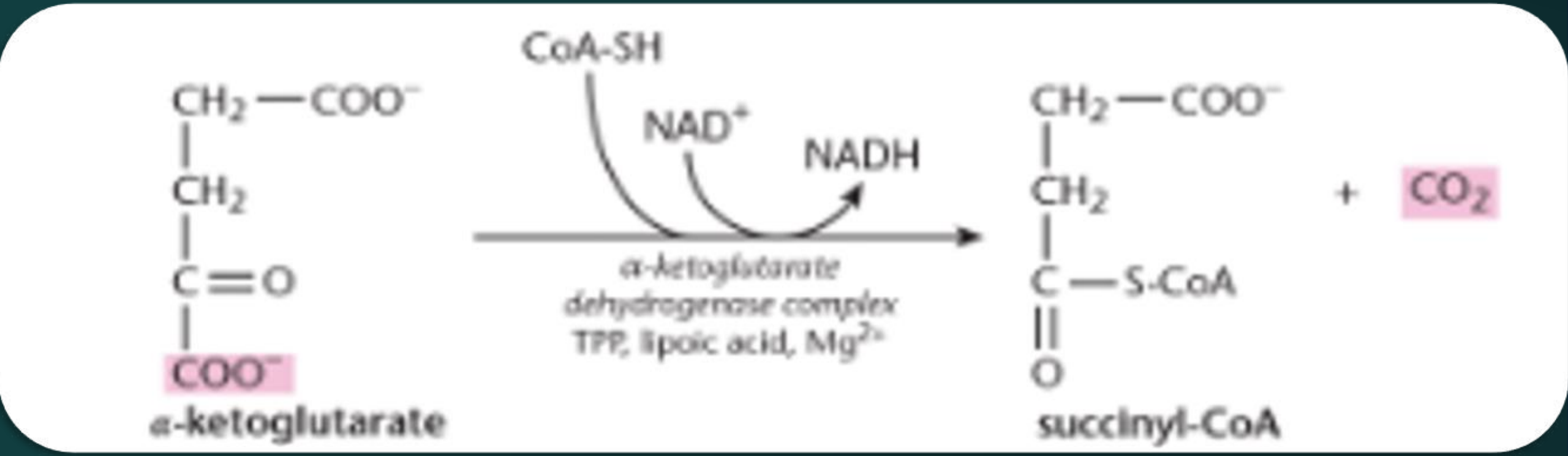
Citric Acid Cycle



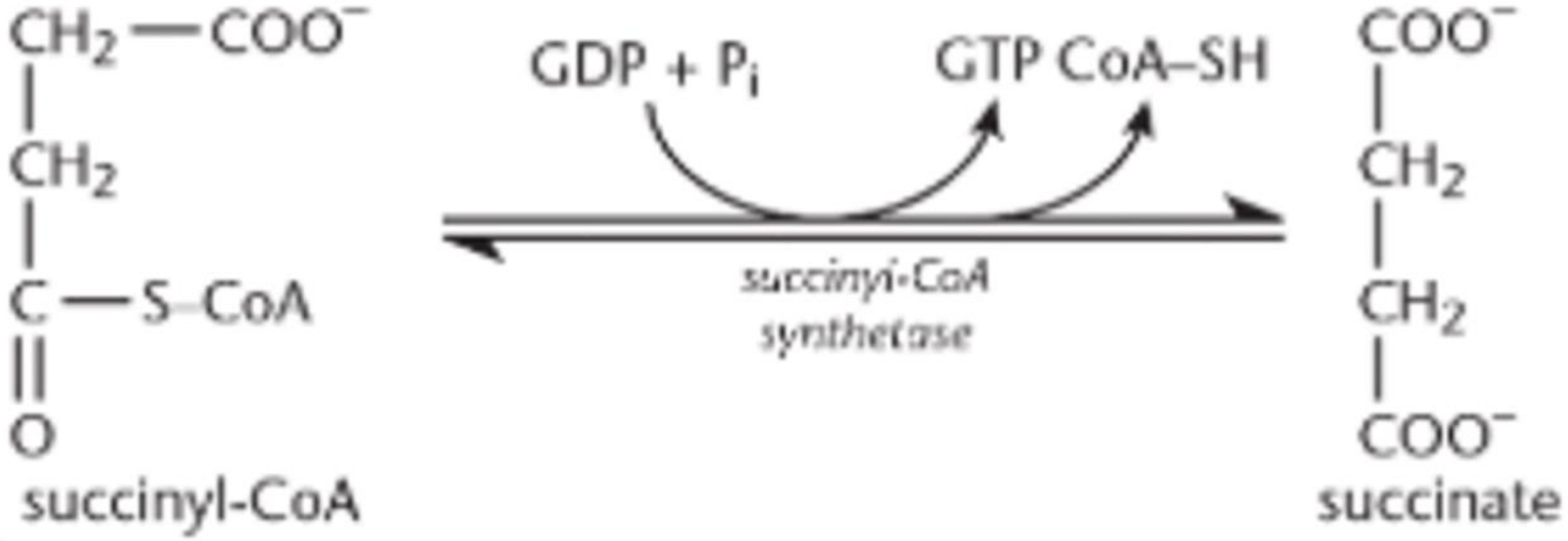
Citric Acid Cycle



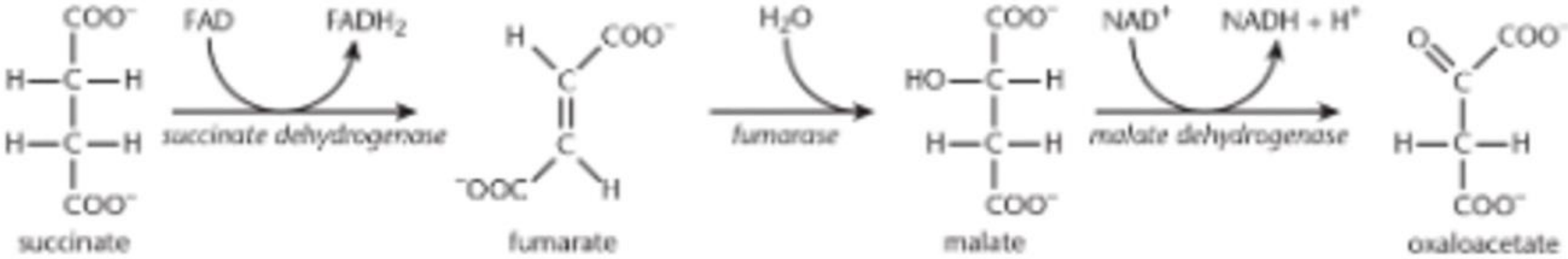
Citric Acid Cycle



Citric Acid Cycle



Citric Acid Cycle



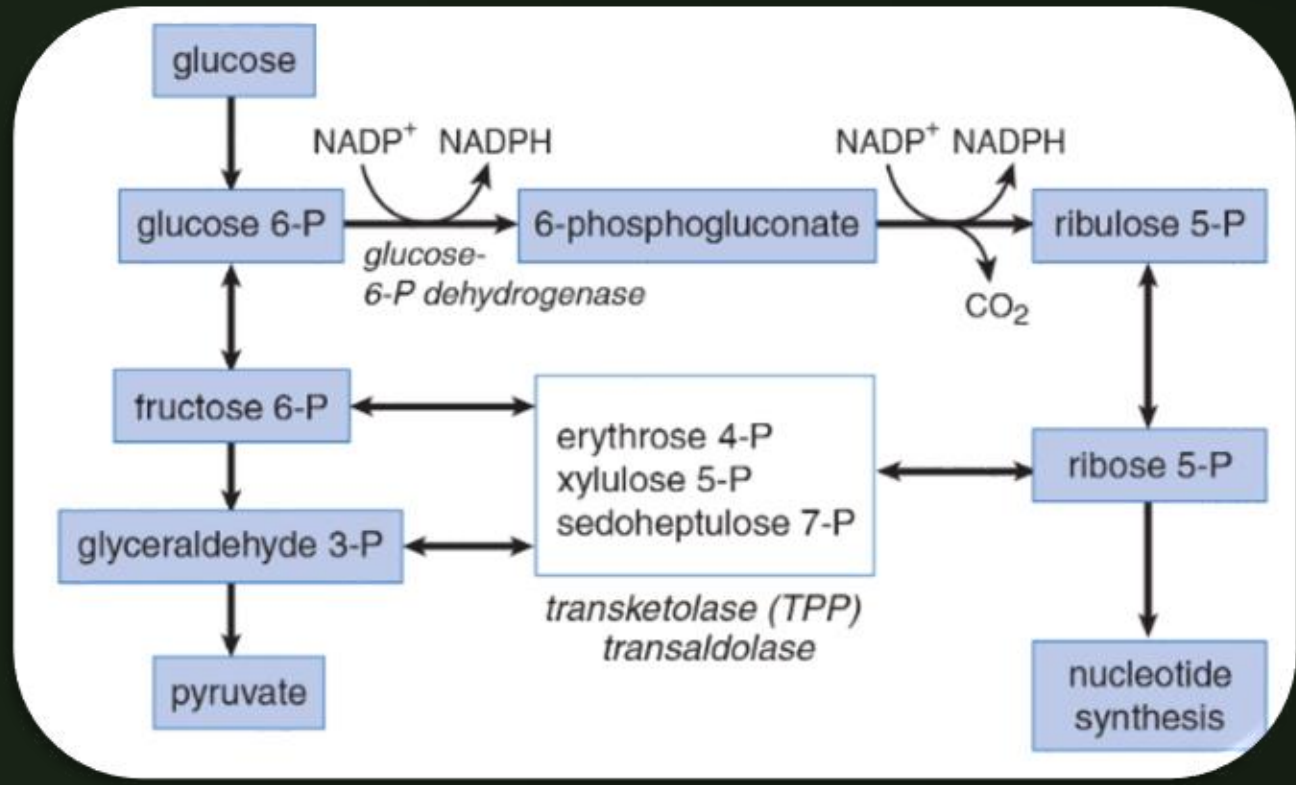
Pentose Phosphate Pathway

Purpose of Pentose Phosphate Pathway

Generation of sugars such as ribose 5-phosphate, which appears in nucleotides and DNA backbones.

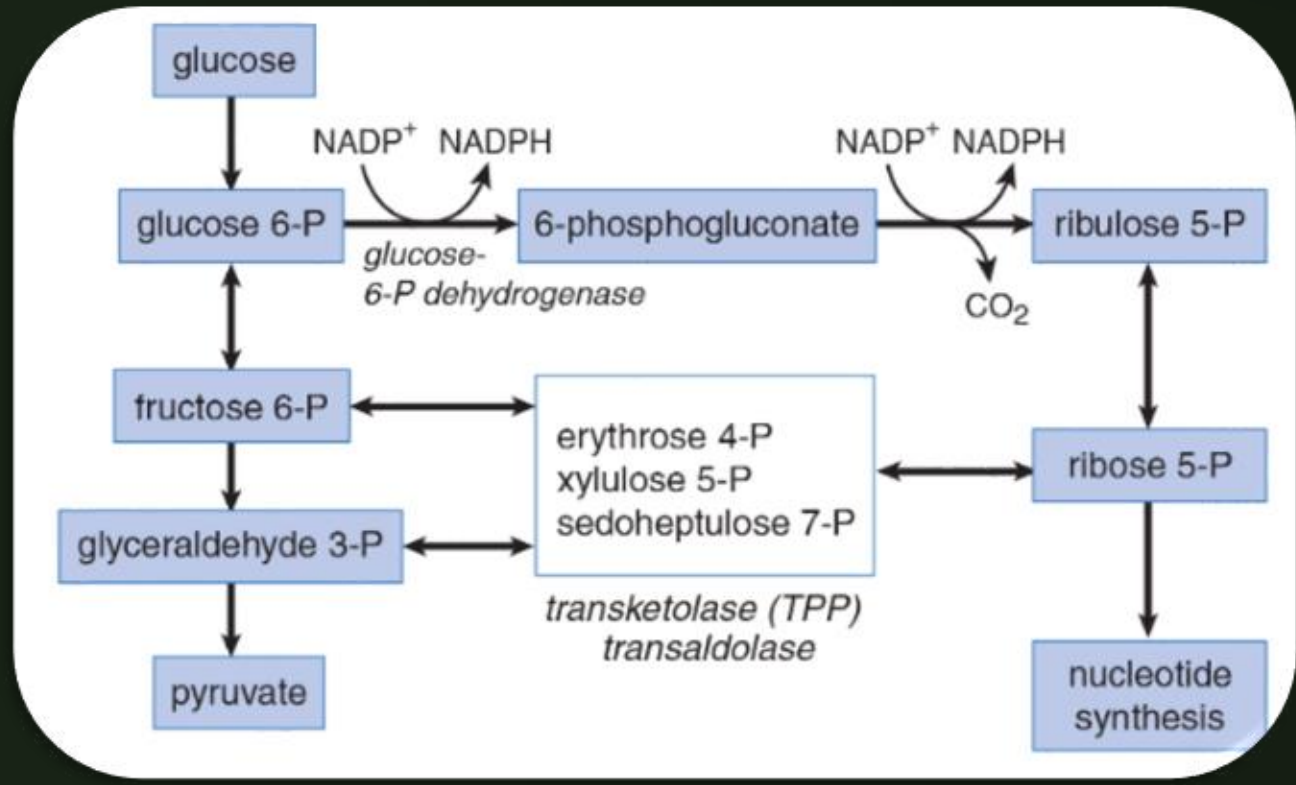
Generation of NADPH. NADPH is similar to NADH, but NADH is used for catabolism, while NADPH is used for anabolic reactions.

The PPP will generate both of these products, and will coordinate with **glycolysis** balance cellular NADPH, ribose 5-phosphate, and ATP needs



Purposes of NADPH in Humans

- Fatty acid biosynthesis
- Cholesterol biosynthesis
- Neurotransmitter biosynthesis
- Nucleotide biosynthesis
- Reduction of oxidized glutathione/antioxidant protection
- Cytochrome P450 monooxygenases
- Production of Cellular Bleach for White Blood Cell Bacteriocidal Activity

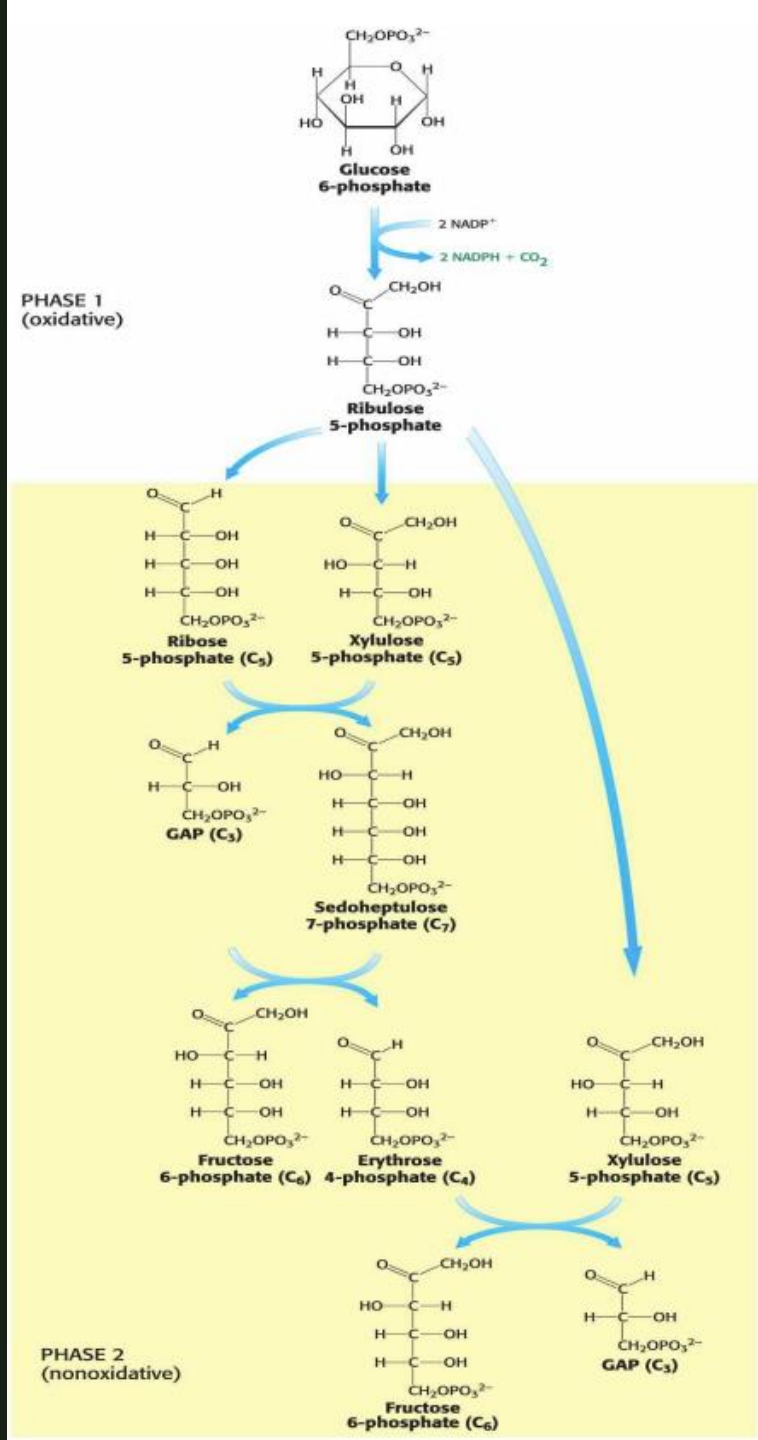


Pentose Phosphate Pathway Stages

The first stage is the Oxidative Phase. This is (obviously) the stage that forms NADPH.

The second/other stage is called the Nonoxidative Phase. This phase uses isomerases and transferases to generate a variety of different-sized sugars.

Both stages branch off of glycolysis. Two entryways allow for versatility.

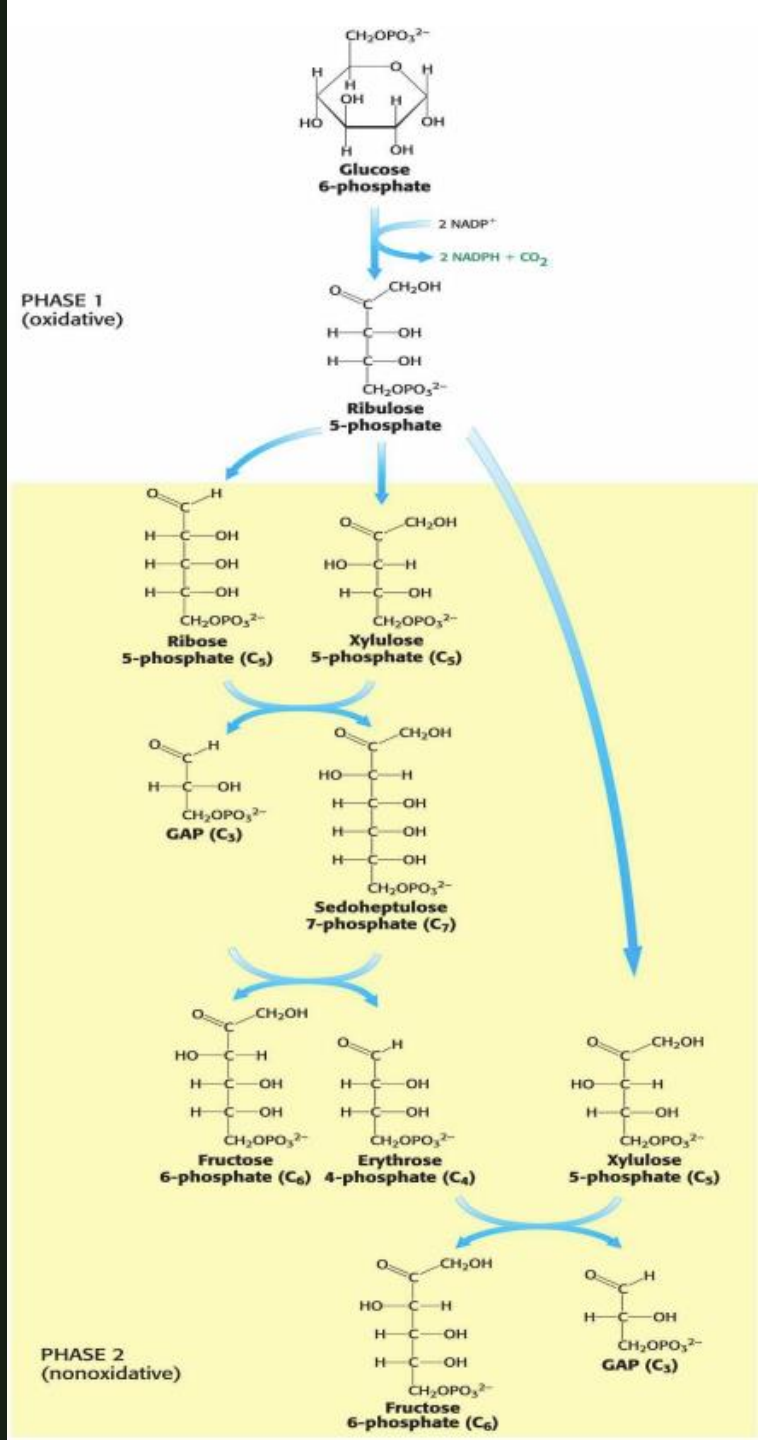


NOTE

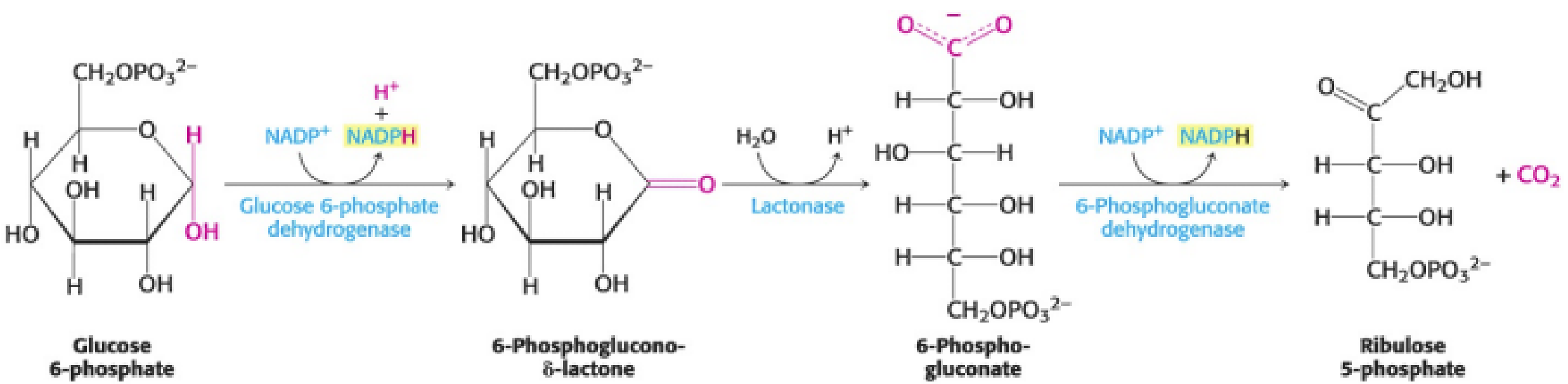
The sugar names are a LOT of clutter/memorization that you do NOT need in your life. Here is what's important:

Remember Ribose, GAP, and Fructose because they appear elsewhere. All sugars have a phosphate on the bottom, largest-numbered carbon. So, the number describes how many carbons are in each sugar

Ex. Erythrose 4-Phosphate is a 4-carbon sugar (tetrose)



Oxidative Stage of PPP



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G6P is turned into Ribulose 5-phosphate. 2 NADPH are generated through oxidation, and one CO_2 is released (making the reaction irreversible). Unidirectional.

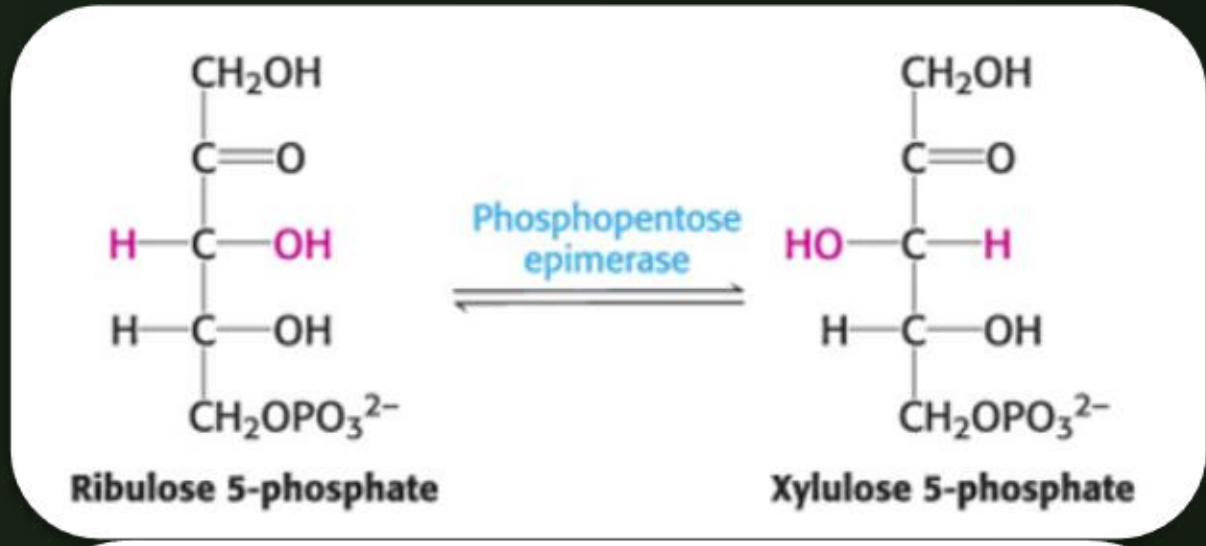
We have a Lactone and an Aldonic Acid (from BCH Chapter 4: Carbohydrates) (Example of Reducing Sugar in action)

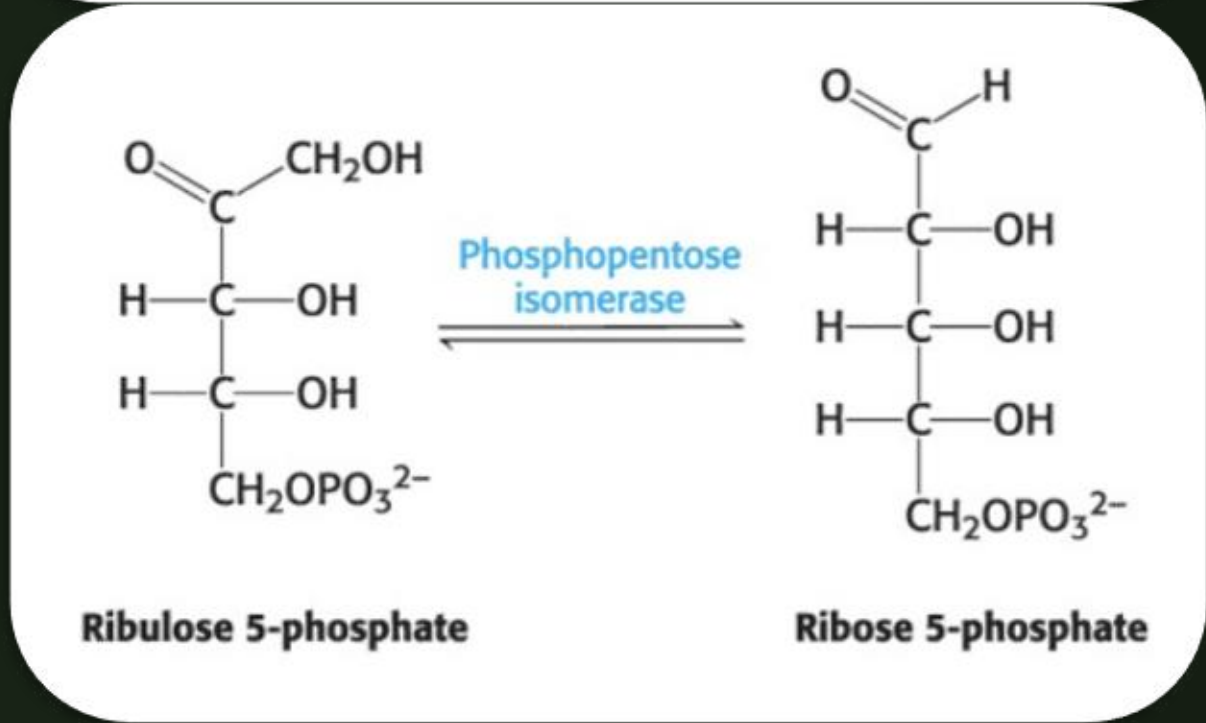
Phosphopentose Epimerase and Isomerase

Phosphopentose epimerase (PPE) converts between ribulose 5-phosphate

Phosphopentose Isomerase converts between Ribulose 5-phosphate and Ribose 5-phosphate

All this means is that our 5-carbon sugars can interconvert amongst each other





Transferases

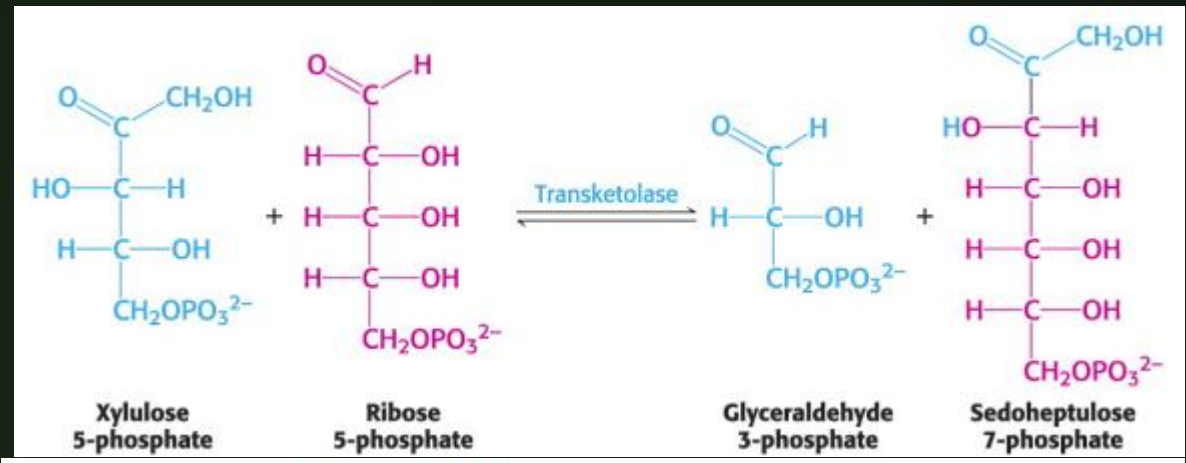
Transketolase moves 2-carbon fragments from the "top" of one sugar to the "top" of another

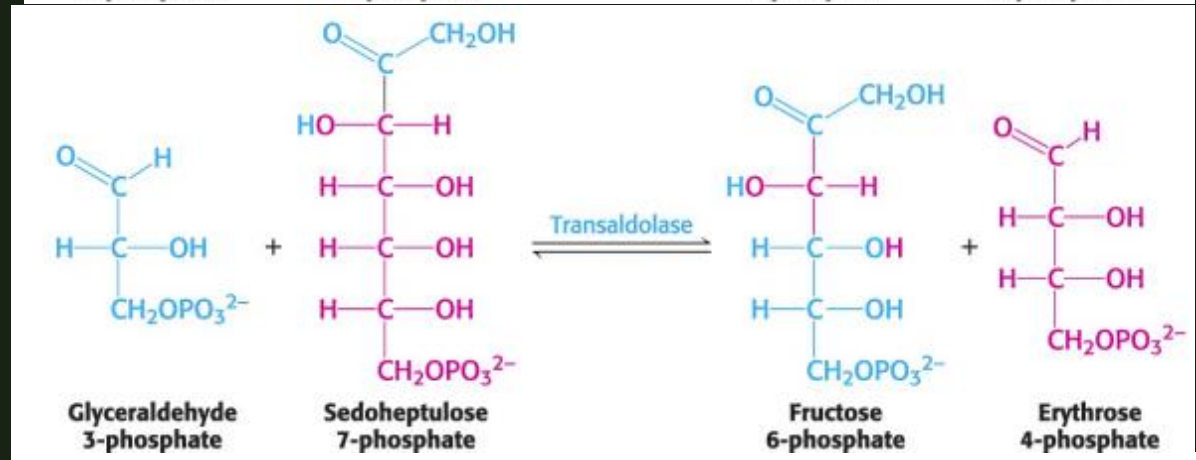
Transaldolase moves 3-carbon fragments from the "top" of one sugar to the "top" of another

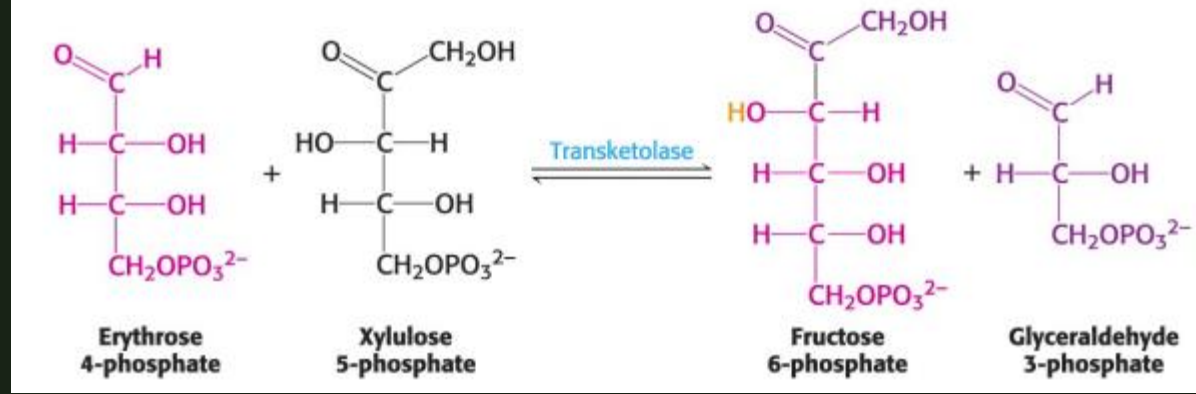
C5 + C5 <-> C3 + C7 <-> C6 + C4,
C4 + C5 <-> C6 + C3

NET: 3C5 <-> 2C6 + C3 = 2 F6P + G3P

All this means is that we ca interconvert from 5-carbon sugars to fructose 6-phosphate and G3P (from/for glycolysis)





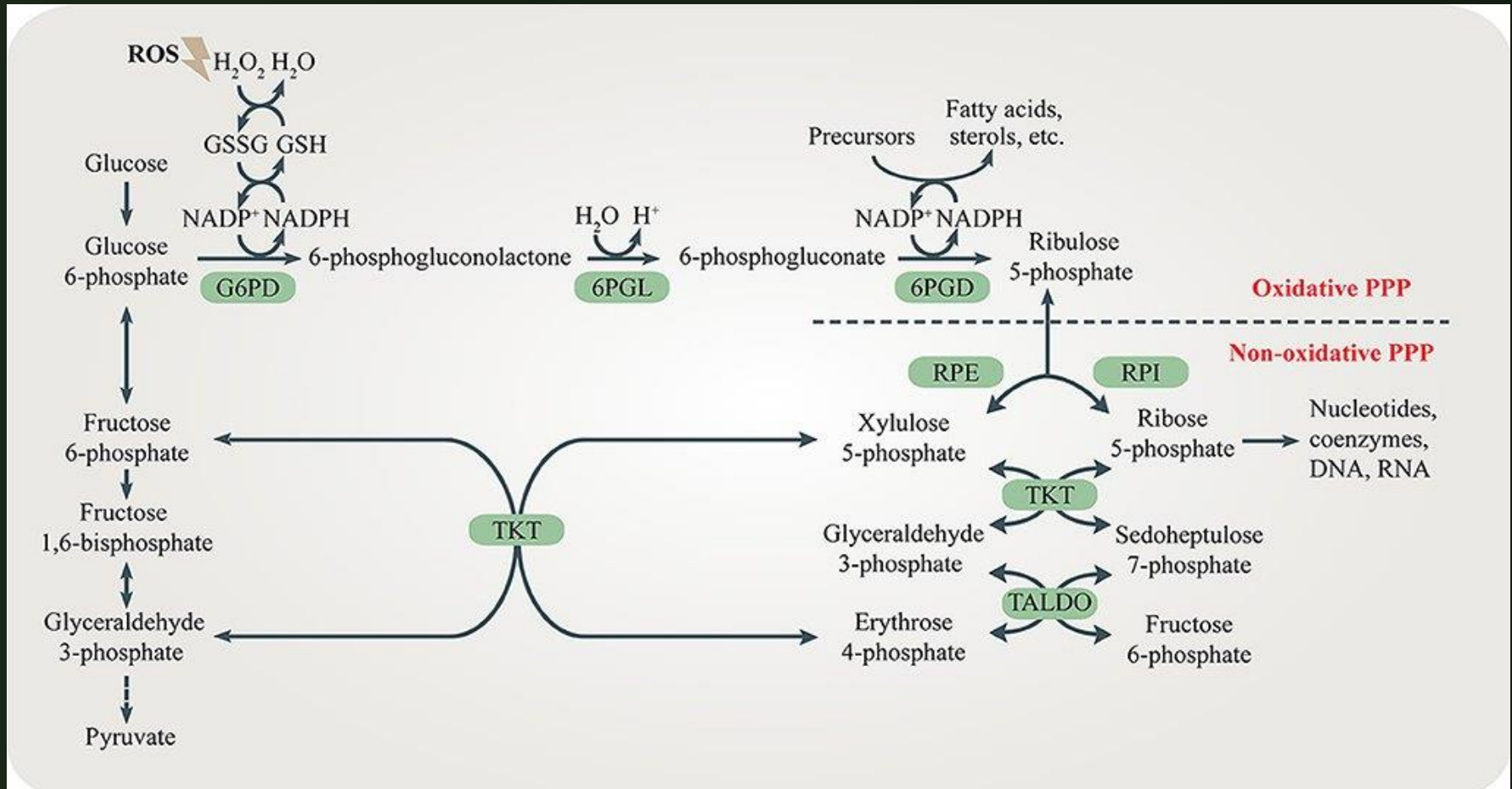


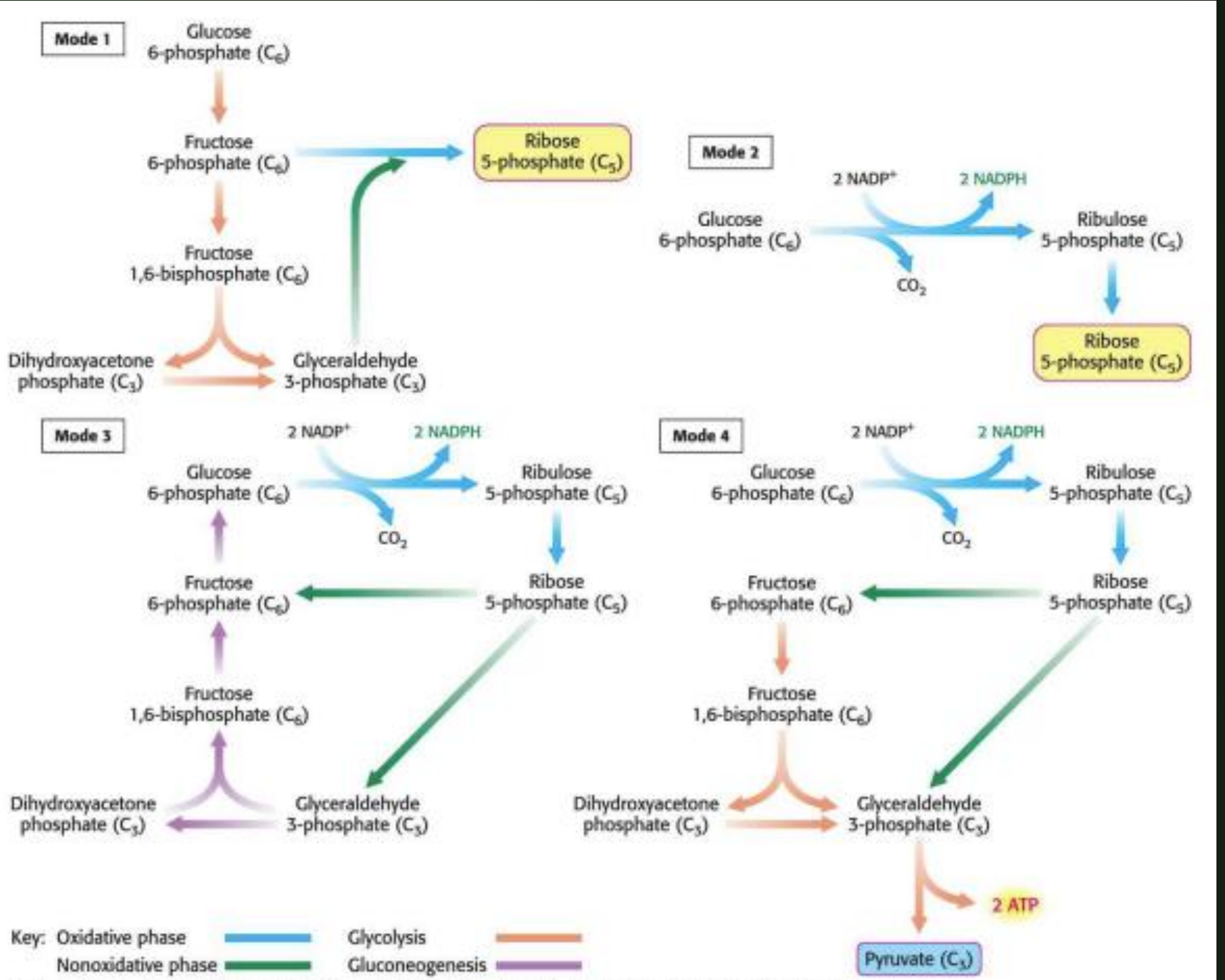
Full Nonoxidative Stage of PPP

The sum of these two processes is the conversion between Ribose 5-Phosphate/Ribulose 5-Phosphate and the intermediates of glycolysis.

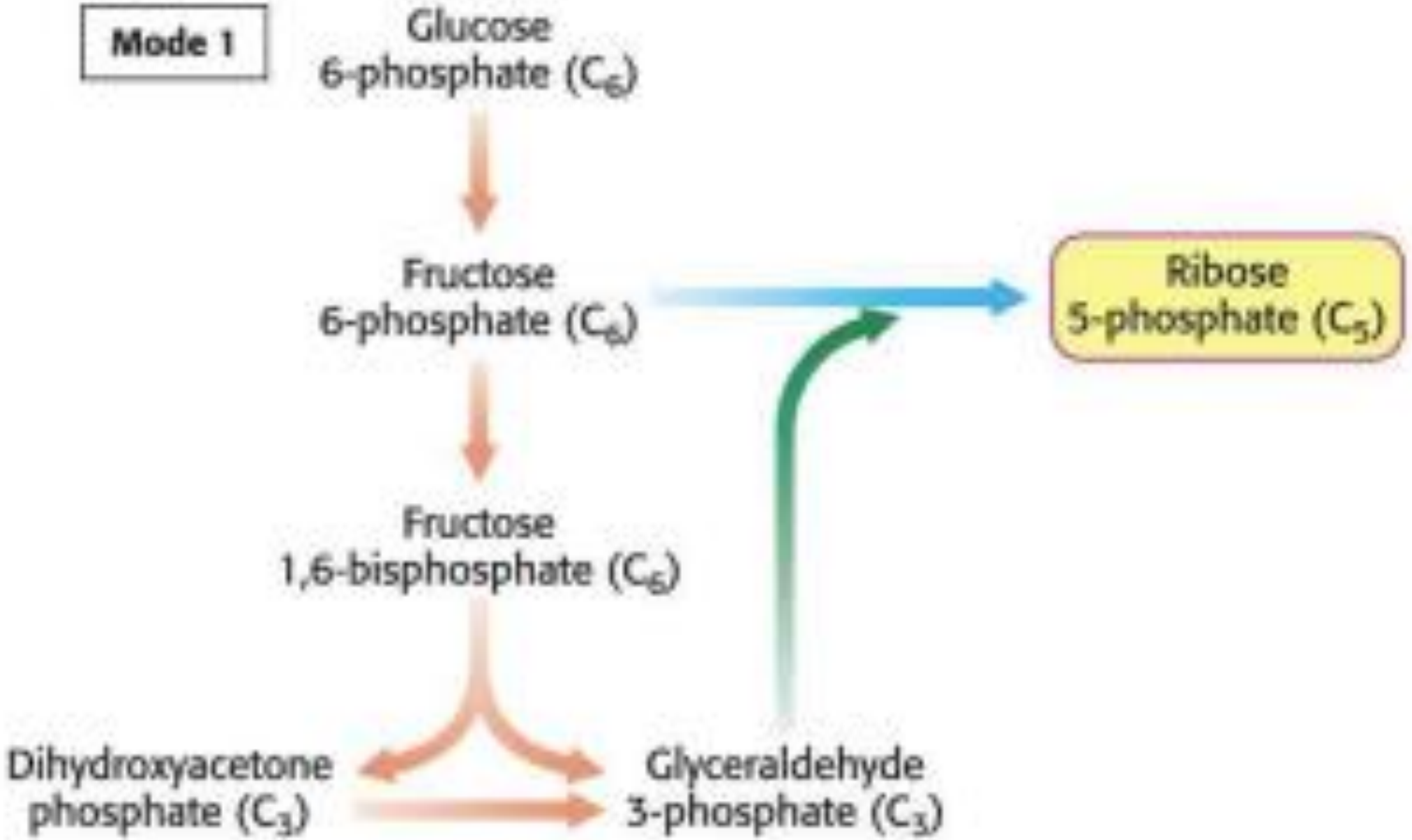
Important: This stage goes forward and backward; the "final" product might be F6P, G3P, R5P, etc.

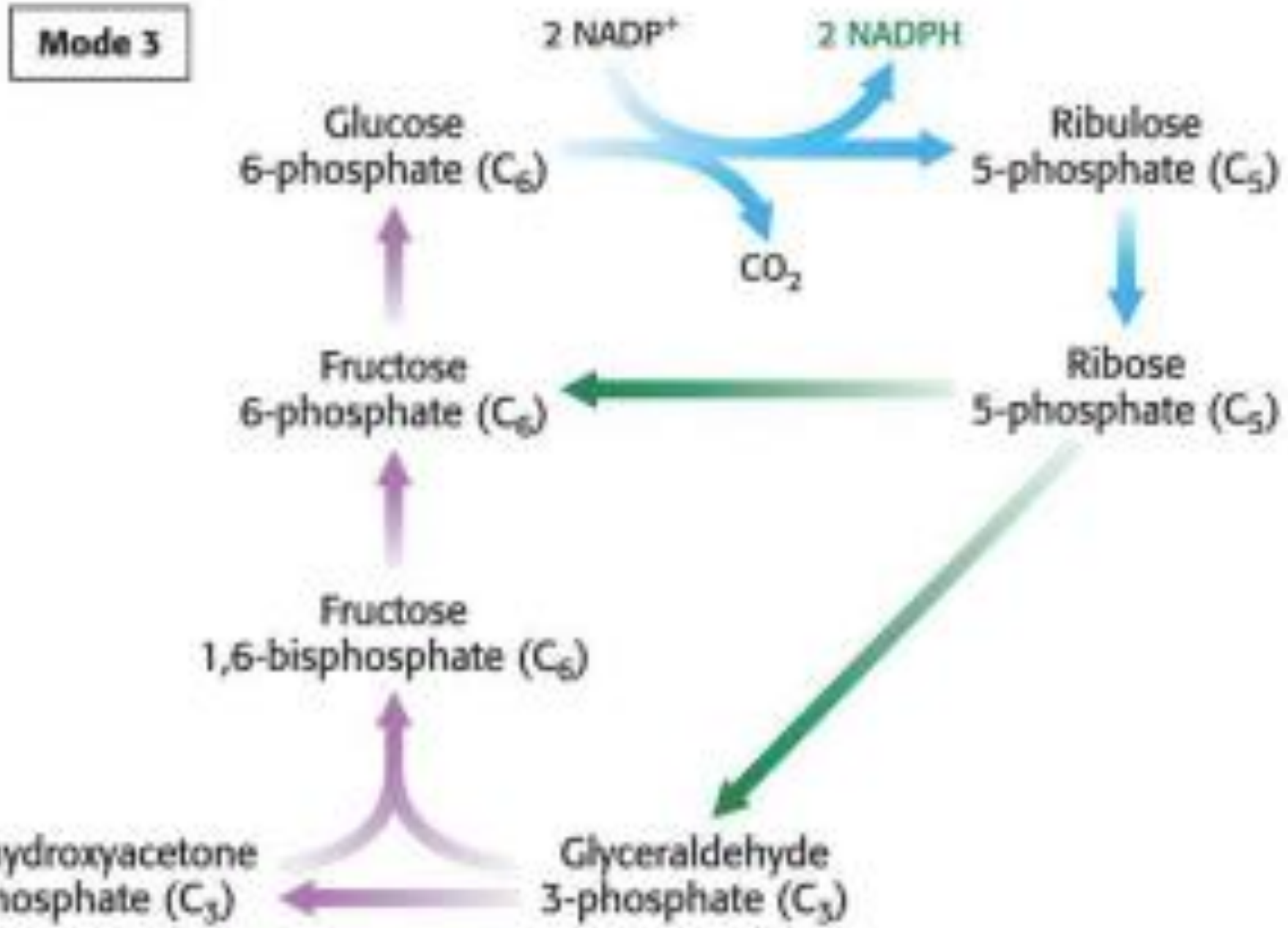
Bidirectionality yields versatility.

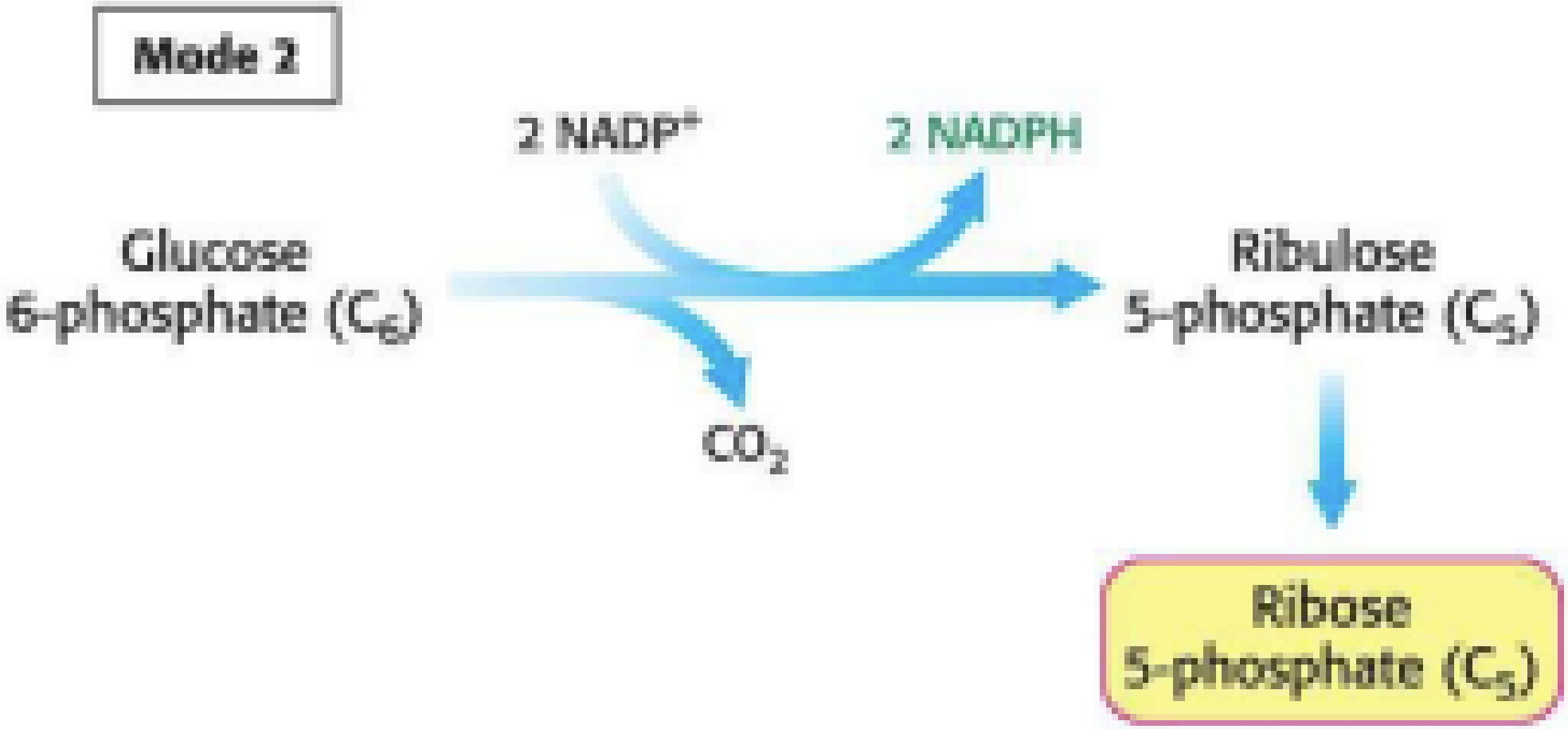


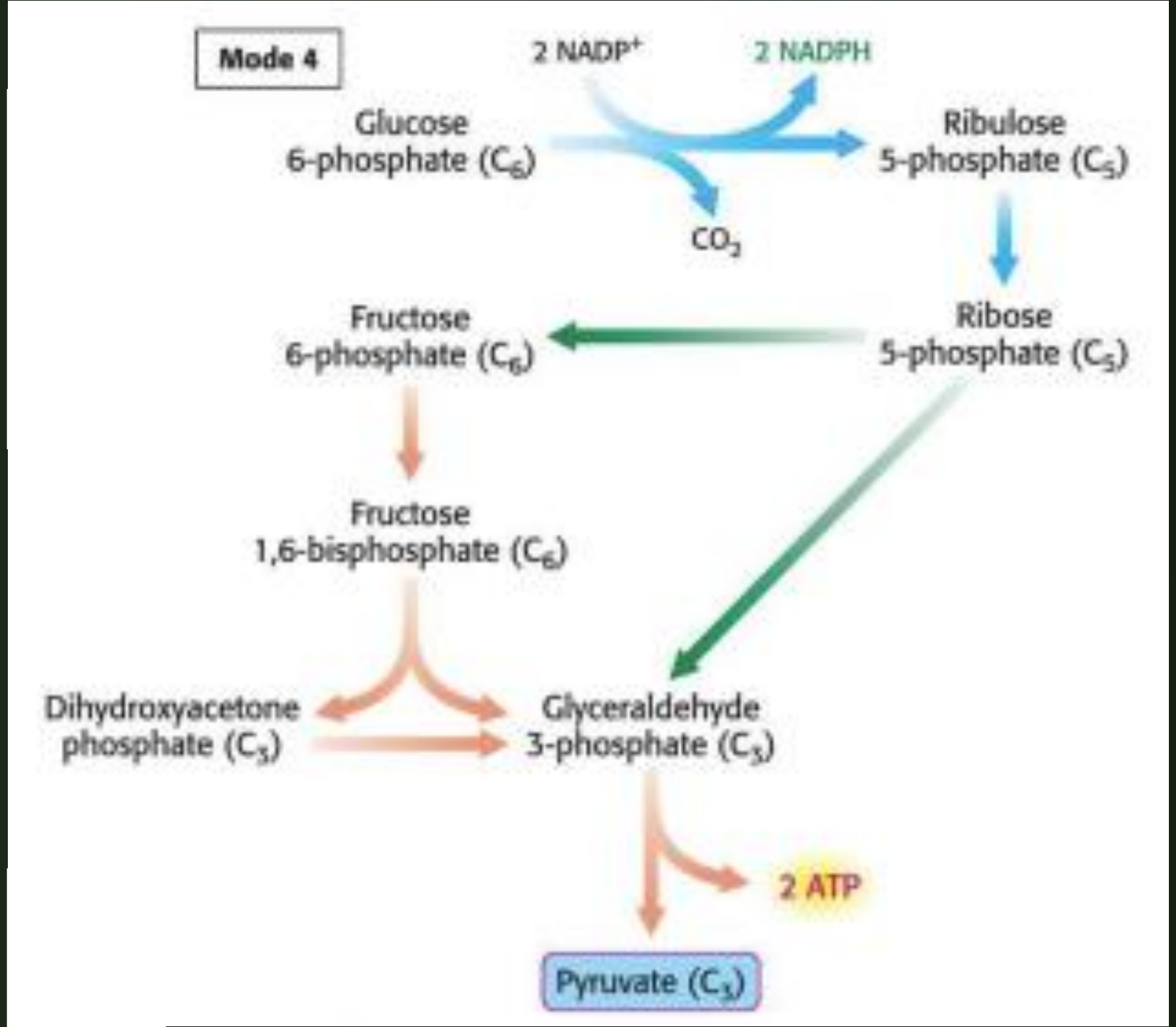


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Mode 1
Glucose 6-phosphate (C_6)
Fructose 6-phosphate (C_6)
Ribose 5-phosphate (C_5)
Fructose 1,6-bisphosphate (C_6)
Dihydroxyacetone phosphate (C_3)
Glyceraldehyde 3-phosphate (C_3)


Mode 3
2 NADP+
2 NADPH
Glucose
6-phosphate (C6)
Ribulose
5-phosphate (C5)
CO2
Fructose
6-phosphate (C6)
Ribose
5-phosphate (C5)
Fructose
1,6-bisphosphate (C6)
Dihydroxyacetone
phosphate (C3)
Glyceraldehyde
3-phosphate (C3)


Mode 2
2 NADP+
2 NADPH
Glucose
6-phosphate (C6)
Ribulose
5-phosphate (C5)
CO2
Ribose
5-phosphate (C5)


Mode 4
2 NADP+
2 NADPH
Glucose
6-phosphate (C6)
Ribulose
5-phosphate (C5)
CO2
Fructose
6-phosphate (C6)
Ribose
5-phosphate (C5)
Fructose
1,6-bisphosphate (C6)
Dihydroxyacetone
phosphate (C3)
Glyceraldehyde
3-phosphate (C3)
2 ATP
Pyruvate (C3)